STOCHASTIC COMPUTER SIMULATIONS

Population dynamics

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1
INTRODUCTION AND USEFUL THEORETICAL RESULTS ON THE GALTON -WATSON PROCESS

Let us consider a Galton-Watson process with mean m and a geometric reproduction law.

$$X_{n+1} = \sum_{i=1}^{X_n} \xi_i \text{ with } X_0 = 1 \text{ and}$$

$$\mathbb{P}(\xi_i = k) = \frac{1}{m+1}\left(\frac{m}{m+1}\right)^k$$

The study of the generating function of this distribution is very useful :

$$\mathbb{E}(s^{X_n}) = f \circ f \cdots \circ f(s) = f^{(n)}(s) \text{ with } f(s) = \mathbb{E}(s^{X_1})$$

Since the distribution is geometric, our generating function is written this way :

$$f(s) = \frac{1}{m+1}\frac{1}{1-\frac{sm}{m+1}} = \frac{q}{1-ps} \text{ with } q = \frac{m}{m+1} = 1-p$$

and

$$\frac{1}{1-f(s)} - \frac{1}{m(1-s)} = 1$$

Iterating this expression gives

$$\begin{aligned}\frac{1}{1-f^{(n)}(s)} &= 1 + \frac{1}{m} + \cdots + \frac{1}{m^{n-1}} + \frac{1}{m^n(1-s)} \\ &= \begin{cases} \frac{m^n-1}{m^n-m^{n-1}} + \frac{1}{m^n(1-s)} & \text{si } m \neq 1 \\ n + \frac{1}{1-s} & \text{si } m = 1 \end{cases}\end{aligned}$$

We only need to evaluate this function in 0 to have an exact expression of : $\frac{1}{1-\mathbb{P}(X_n=0)} = \frac{m^{n+1}-1}{m^{n+1}-m^n}$ if $m \neq 1$ and then $\mathbb{P}(X_n > 0) = m^n \frac{1-m}{1-m^n}$

Considering the independence of the offsprings , the law of each generation can always relate to the case with $pop := X_0 = 1$ starting with only one person.

More precisely, starting with i individuals, $X_n = \sum_{k=1}^{i} X_{n,k}$, $X_{n,k}$ being the number of people who descend from kth first person. Thus, we only need the convolution product of the original ($pop = 1$) i times to find the law with $pop = i$. With $X_0 = i$ the probability of extinction is then $\left(\frac{m^{n+1}-1}{m^{n+1}-m^n}\right)^i$ if $m \neq 1$.

Another classical aspect concerns the behaviour of super critical G.W. processes for whom every state other than 0 and ∞ is transient. Moreover $\mathbb{P}(X_\infty = 0) = 1 - \mathbb{P}(X_\infty = \infty) = 1/m$ as the smallest root of the equation $f(s) = s$ in $[0, 1]$ [Mél].

Some other theoretical results found in the literature are used in this project:

- $$\mathbb{P}(X_n = j \mid X_n > 0, X_0 = i) \xrightarrow[n\to\infty]{} (1-m)m^{j-1}$$

  is called the Yaglom limit.

- $\frac{X_n}{m^n}$ converges almost surely Kersting

A more complex reproduction law m_0, m_1, p is the probability over the environment ie the probability that the reproduction law at a given generation is a geometric law of mean m_1. Let $f_1(s) = \frac{p_1}{1-q_1 s}$ and $f_2(s) = \frac{p_2}{1-q_2 s}$ respectively.

We have $\frac{1}{1-f_i(s)} - \frac{1}{m_i(1-s)} = 1$, so $\frac{1}{1-f_i(f_j(s))} = 1 + \frac{1}{m_i} + \frac{1}{m_i m_j(1-s)}$ thus

$$\mathbb{E}(s^{X_n} \mid f_{i_0}, ..., f_{i_{n-1}}) = 1 - \frac{1}{1 + \frac{1}{m_{i_0}} + ... + \frac{1}{m_{i_0}...m_{i_{n-2}}} + \frac{1}{m_{i_0}...m_{i_{n-1}}}\frac{1}{1-s}}$$

In particular,

$$\mathbb{P}(X_n = 0 \mid f_{i_0}, ..., f_{i_{n-1}}) = 1 - \frac{1}{1 + \frac{1}{m_{i_0}} + ... + \frac{1}{m_{i_0}...m_{i_{n-2}}} + \frac{1}{m_{i_0}...m_{i_{n-1}}}}$$

where $p_1 = p$ and $p_0 = 1 - p$

Finally,

$$\mathbb{E}(s^{X_n}) = \sum_{(i_0,...,i_{n-1})\in 0,1^n} p_{i_0}...p_{i_{n-1}}(1 - \frac{1}{1 + \frac{1}{m_{i_0}} + ... + \frac{1}{m_{i_0}...m_{i_{n-2}}} + \frac{1}{m_{i_0}...m_{i_{n-1}}}\frac{1}{1-s}})$$

$$\mathbb{P}(X_n = 0) = \sum_{(i_0,...,i_{n-1})\in 0,1^n} p_{i_0}...p_{i_{n-1}}(1 - \frac{1}{1 + \frac{1}{m_{i_0}} + ... + \frac{1}{m_{i_0}...m_{i_{n-2}}} + \frac{1}{m_{i_0}...m_{i_{n-1}}}})$$

In the case of multiple environments , some results concern the speed of convergence of the probability of survival:

- if $\mathbb{E}(\log m(p)) < 0$ the process is subcritical and converges almost surely to 0, and more complex asymptotics are described in [JL03], [PE 72].
- if $\mathbb{E}(\log m(p)) = 0$ the process is called critical and converges almost surely to 0 more slowly $(n^{\frac{-1}{2}}$
- if $\mathbb{E}(\log m(p)) > 0$ the process is over-critical.
- The Kersting lemma generalizes to $\frac{X_n}{\prod_{i<n} m(p_i)}$

2
IMPORTANCE SAMPLING AND APPLICATIONS

2.1 Probability of survival

2.1.1 General Outlook

Looking at the aforementioned expression of the probability of survival we see that $p = \mathbb{P}(X_n > 0) < m^n$ in the case of $m < 1$ which means that the probability decreases exponentially fast . For instance, for $m = 0.1$ we are already under 10^{-10}. If we were to use the naive Monte- Carlo approach with $N = 10^4$ simulations the standard deviation would be of around $\frac{\sqrt{p(1-p}}{\sqrt{N}} \simeq 10^{-7}$ which is 1000 times higher than the targeted value. Equivalently, we can say that we need 10^{12} simulations to have an estimation with a standard deviation 10 times smaller than the estimated value.

This showcases the necessity of a more advanced approach.Thus we start by testing the Importance Sampling method. We first notice that we are a given a family of Markov chains from the start, one defined for each m. We also point out that all the states can be reached by all the Markov chains ($\forall m' P_{m'}$ dominates P_m)

Before delving deeper into the method, we should find a way to compute in a precise way the transition kernels of those Markov chains, in order to find the likelihood ratio of a trajectory under the initial and changed probability.

$$\frac{\mathbb{P}(X_1, X_2, ..., X_n)}{\mathbb{Q}(X_1, X_2, ..., X_n)} = \frac{\prod_i^{n-1} P(X_i, X_{i+1})}{\prod_i^{n-1} Q(X_i, X_{i+1})}$$

We then noticed that given i the population at time n, the population at time $n+1$ follows a negative binomial law of parameters $(i, \frac{1}{m+1})$, i.e a sum of i random variables each following a geometric law with $1/(m+1)$ the probability of its success. That is very useful since it helps us to compute exactly the likelihood ratio of a given trajectory without having to approximate it too. If we simulate the Galton-Watson process with m', we can estimate the expectancy of a certain event $\mathcal{A}$ happening under the m Markov chain with the relation

$$\mathbb{E}_m(\mathcal{A}) = \mathbb{E}_{m'}\left(\mathcal{A}\frac{\prod_i^{n-1} P(X_i, X_{i+1})}{\prod_i^{n-1} Q(X_i, X_{i+1})}\right) \simeq \frac{\sum_{i=0}^{L} \mathbb{1}_{X^{(i)} \in \mathcal{A}} \left(\frac{\prod_k^{n-1} P(X_k^{(i)}, X_{k+1}^{(i)})}{\prod_k^{n-1} Q(X_k^{(i)}, X_{k+1}^{(i)})}\right)}{L}$$

We should then choose m' such that

$$Var_{P_{m'}}\left(1_{X_n > 0} \prod \frac{P(X_i, X_{i+1})}{Q(X_i, X_{i+1})}\right) \text{ is minimal.}$$

2.1.2 Choosing the right probability change

The idea is then to make the rare events not so rare by simulating them under another Markov chain, and then to correct this bias by multiplying by the likelihood ratio. Thus, choosing a smaller $m' < m$ would clearly be of no use. Choosing a very big $m' > 1$ can be problematic too, as it makes the trajectories very high which in turn make the likelihood ratio very low. It can easily create a loss

of precision due to very low values and the decimal encoding. It also has a high variance due to wildly varying likelihoods, over different orders of magnitude which makes the computation of the mean hard as it tries to sum numbers that are orders of magnitude apart. This entails a loss of precision numerically.

Here is for the case $m = 0.7$ and $n = 10, pop = 1$ which gives a probability of 8.10^{-3} (a not too rare event) an optimization process that iterates over different values of m' trying to find the best one:

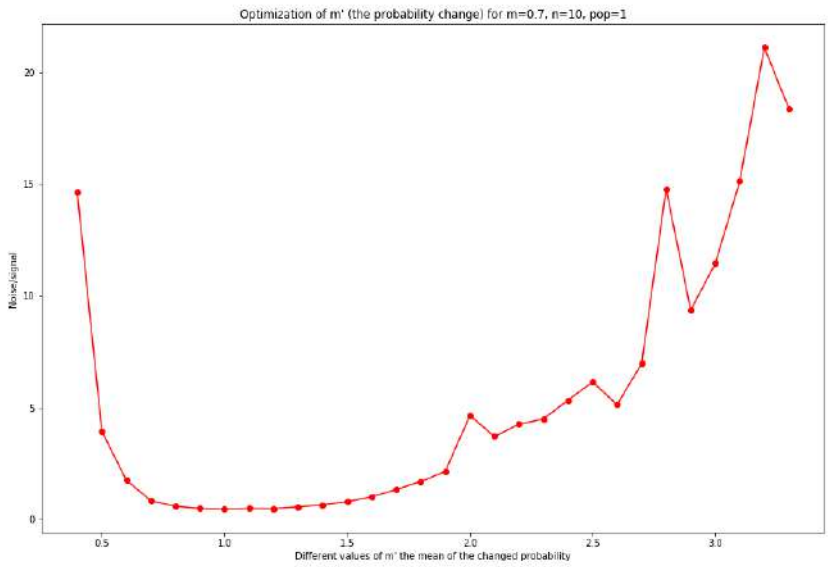

Figure 1: Noise/Signal optimization criteria for different probability changes
m' optimisation for $m = 0.7$

The noise over signal ratio of the naive 100 Monte Carlo estimation for the case $m = m' = 0.7$ is 1.06, which is roughly 2.5 times more than the 0.45 noise over signal for the m=1. Our $m = 1$ initial heuristic was confirmed thus in the case of not so rare events graph 1. Increasing the number N of samples divides naturally the noise by a factor $\sqrt{N}$. The importance sampling method is not the most adapted for the not so rare events, because the likelihood ration takes time to be computed.

Can we confirm it in the case of rarer events? For instance, we have tried it with $m = 0.1$ which gives a probability of survival after 10 generations of the order of 10^{-10}.

Even though the graph 2 is less smooth, we see that the changing of probability with the case $m = 1$ still yields relatively good results. We will now fix $m' = 1$ and try to estimate the probability for several m in the following graph.

2.2 Probability of exceeding resources:

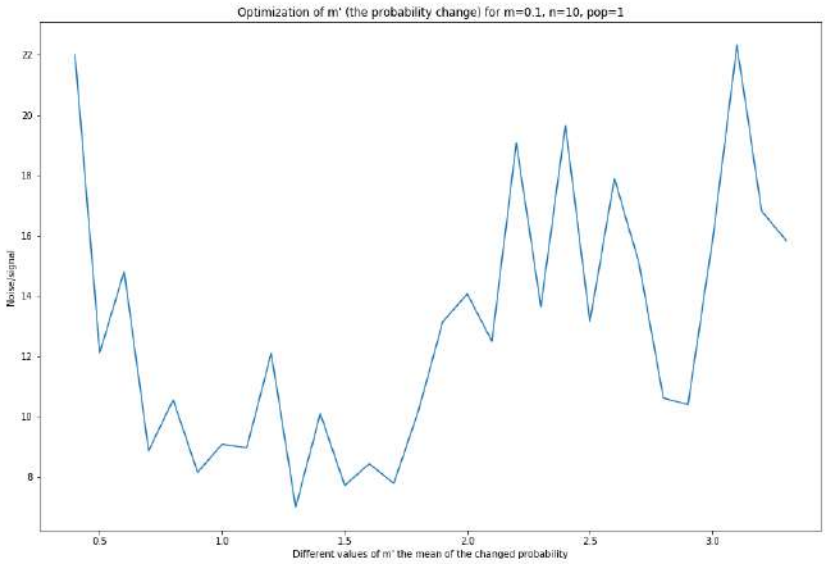

Figure 2: Noise/Signal optimization criteria in the case of a rare event for different probability changes m' optimization for $m = 0.1$

2.2.1 Deterministic method:

In order to determine the probability of exceeding a given level x of resources $P(max(X_i) > x)$, we try to compute first the $P(max(X_{i\leq n}) < x) = P(X_1 < x, X_2 < x, ..., X_n < x)$. We notice that if

$$P_n = \begin{pmatrix} p_{0,n} \\ p_{1,n} \\ \vdots \\ p_{x,n} \end{pmatrix}$$

With $p_{i,n} = P(X_n = i)|max(X_{i<n}) < x)$ We easily see then that:

$$P_{n+1} = TP_n = \begin{pmatrix} t_{0,0} & t_{0,1} & \cdots & t_{0,x} \\ t_{1,0} & t_{1,1} & \cdots & t_{1,x} \\ \vdots & \vdots & \ddots & \vdots \\ t_{x,0} & t_{x,1} & \cdots & t_{x,x} \end{pmatrix} \begin{pmatrix} p_{0,n} \\ p_{1,n} \\ \vdots \\ p_{x,n} \end{pmatrix}$$

with $t_{i,j}$ the probability of going from j to i equals the probability mass function of a negative binomial of parameters $(j, 1/(m+1)$ evaluated on i . Using `nbinom.pmf(i,j, 1/(m+1))` gives a deterministic estimation of the $P(max_{i\leq n}(X_i) > x)$.

In order to have an estimation of $\mathbb{P}(max_{i\in\mathbb{N}}(X_i) > x) = \mathbb{P}(\bigcup_{n\in\mathbb{N}} max_{i\leq n}X_i) = \lim \mathbb{P}(max_{i\leq n}(X_i) > x)$

The same can be done for $P(\sum^n X_i > x)$ with the only difference being that the transition probability matrix has at the column j and line $j+i$ the probability of i individuals having j individuals which is `nbinom.pmf(i,j, 1/(m + 1))`, in order to account for the the fact that we sum at each iteration.

2.2.2 Choosing the right probability change for the importance sampling:

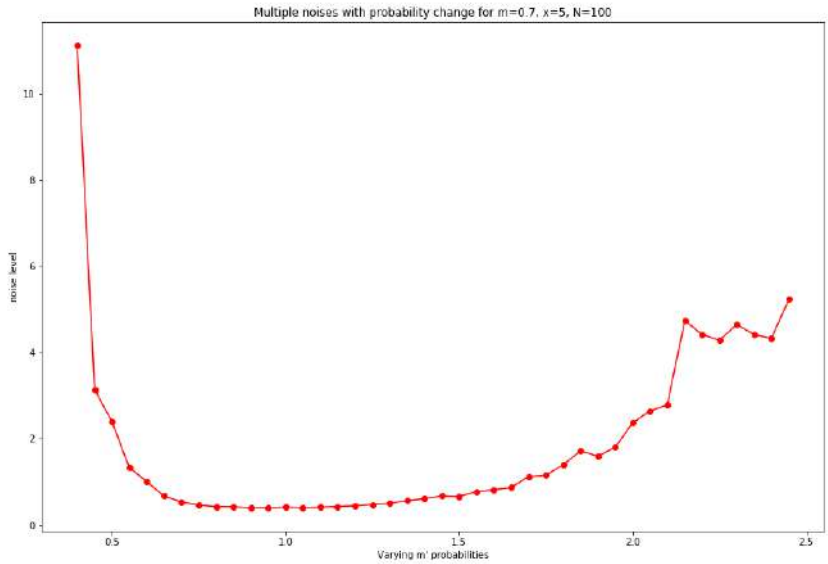

Figure 3: Noise/Signal optimization criteria for different probability changes for x=5, m=0.7 m' optimization for $m = 0.7, x = 5$

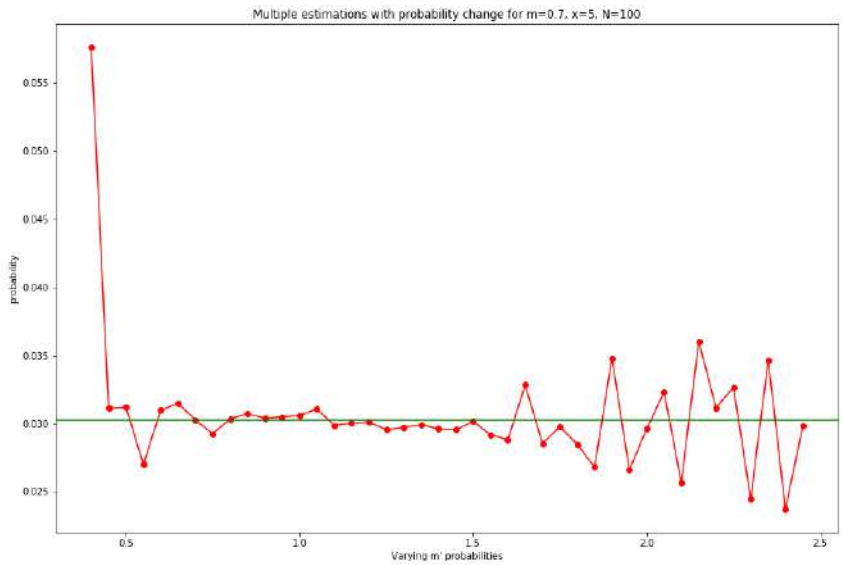

Figure 4: Estimated probability for different probability changes m' for x=5, m=0.7 m' optimization for $m = 0.7, x = 5$

Doing the same as in the previous section we find that the parameter the probability change translates a balance between the events actually happening and a not too small likelihood ratio of the trajectories. Figures 4 and 3 show that the optimal m' is here as well around 1. Another heuristic might also be to set m' so that $E(X_n) = m'^n = x$. i.e $m' = \sqrt[n]{x}$.

For $x = 30$ the event of an $m = 0.7$ process exceeding x is very rare and still not so indistinguishable . For the event of exceeding $x = 30$, we plotted again the noise/estimation ratio of different importance sampling estimators. We notice here that the values around $1.4 \simeq 30^{0.1}$ give a less acceptable result (figures 5, 6) than the simple $m' = 1$. A possible explanation to that is the very low values taken by the likelihood ratio.

For higher x, even the importance sampling method gets noisy. Here are the same graphs as before but for $x = 30$

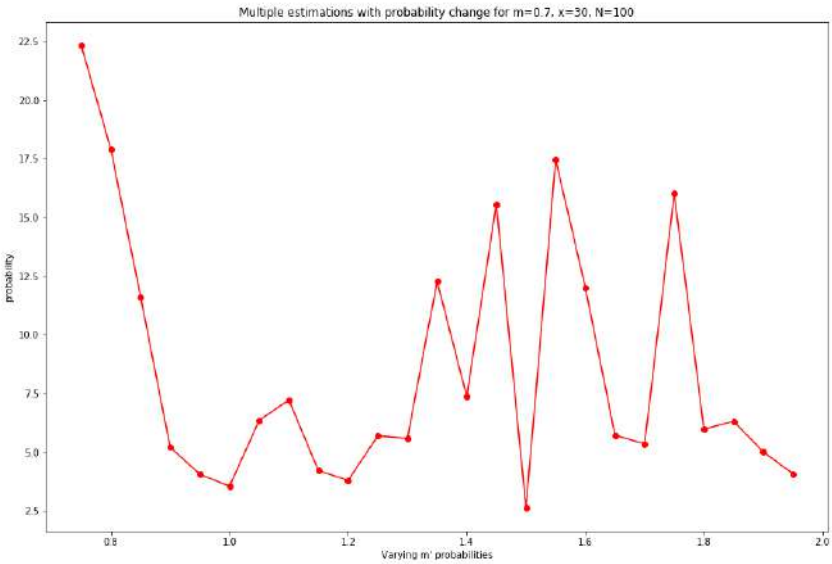

Figure 5: Noise/Signal optimization criteria for different probability changes for x=5, m=0.7 m' optimization for $m = 0.7, x = 30$

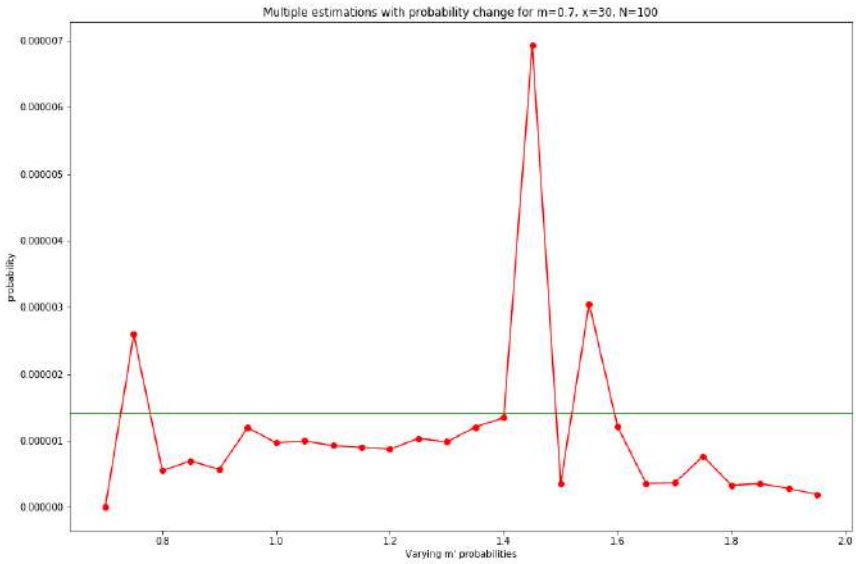

Figure 6: Estimated probability for different probability changes m' for x=5, m=0.7
m' optimization for $m = 0.7, x = 30$

2.2.3 Illustrating some theoretical results

Using the importance sampling method with $m' = 1$ and n_0 such that $P(X_{n_0} > 0) = 10^{-6}$ gives us a very good approximation (graph 7) of the limit, *i.e* $P(\max_{i<n_0} X_i > x) \simeq P(\max_{i\in\mathbb{N}}(X_i) > x)$, the error we will be making from the remaining gap will be less that 10^{-6} considering that the already dead trajectories have no chance whatsoever to exceed the threshold.

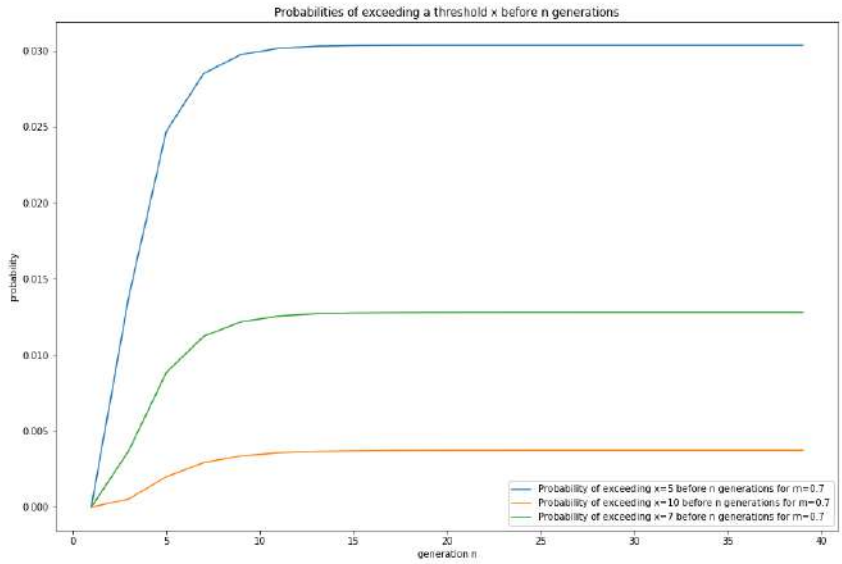

Figure 7: Probability of exceeding x before n generations
Visualizing the convergence of the probabilities

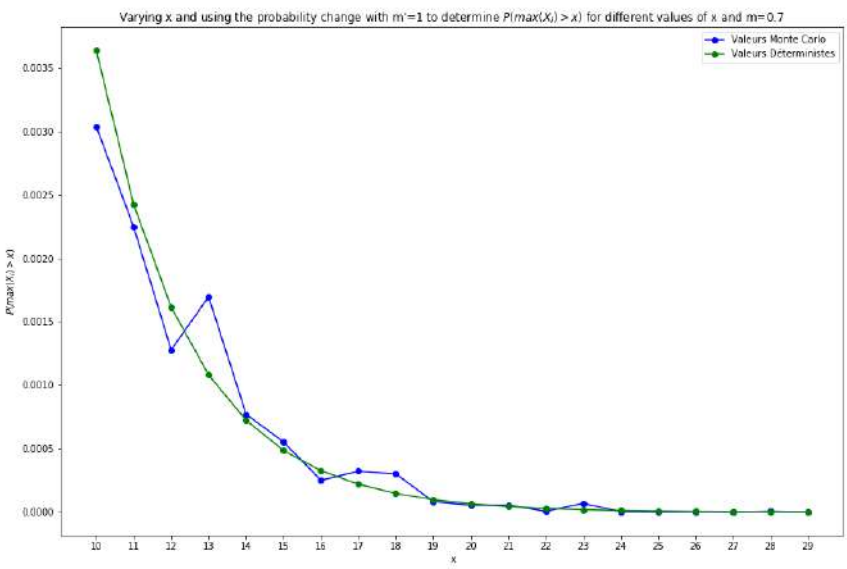

Figure 8: Probability of exceeding x as a function of x for m=0.7 and pop=1 with importance sampling based on m'=1
Evaluation of the importance sampling method on the exceeding resource limit event

It is clear from (figure 9) that in the case of $m = 1$ the speed of convergence to 0 of the probability

of exceeding x is much less than that of the cases in $m < 1$. The article [Ner77] proves that is the case of $m = 1$, $P(\max_{i\in\mathbb{N}}(X_i) > x) \sim \frac{\texttt{pop}}{x}$.

This theoretical result is illustrated in graph 10.

Another theoretical result in [Ner77] states that if we have a regular subcritical Galton-Watson process *i.e* with $f(a) = a$ for an $a > 1$, the speed of convergence in the cases of $m < 1$ is at a^{-x}/x. In particular, for this geometric law of reproduction $a = 1/m$ and thus we can showcase that result in 11.

As we can see in the graph, there is no convergence of $P(\max_{i\in\mathbb{N}}(X_i) > x)m^{-x}x$, however it is the only rate that make those functions verify

$$0 < \liminf_x P(\max_{i\in\mathbb{N}}(X_i) > x)m^{-x}x < \limsup P(\max_{i\in\mathbb{N}}(X_i) > x)m^{-x}x < \infty$$

. The super-critical cases have their probability of exceeding a fixed threshold converging to the probability of them exploding which equals $1 - \frac{1}{m}$ *c.f* the theoretical results of part 1.

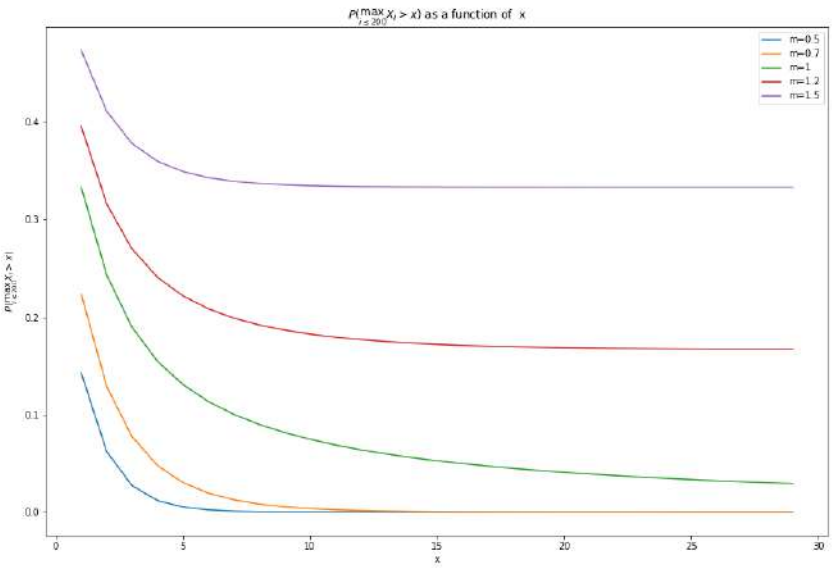

Figure 9: Probability of exceeding x as a function of x for different m
Probability of exceeding convergence

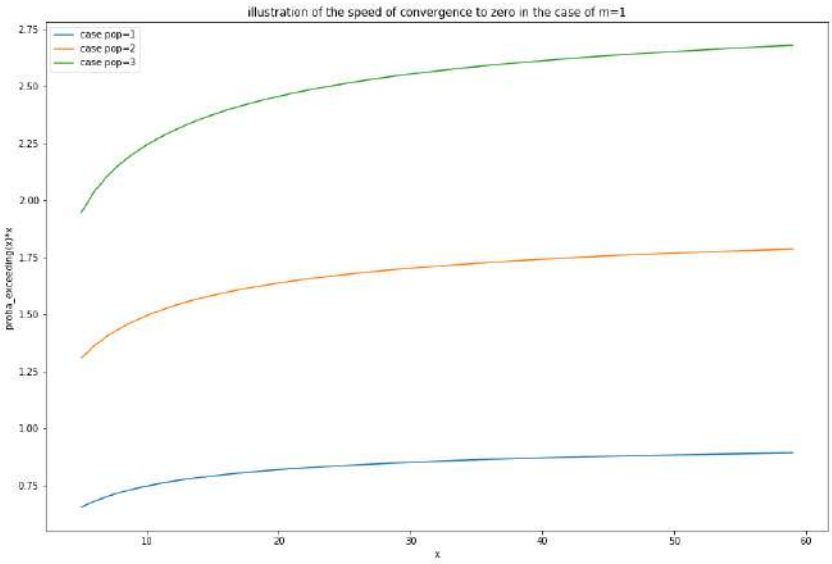

Figure 10: $\mathbb{P}(\max_{i\in\mathbb{N}}(X_i) > x)x$ for $m = 1$ with different pops
Speed of convergence for $m = 1$

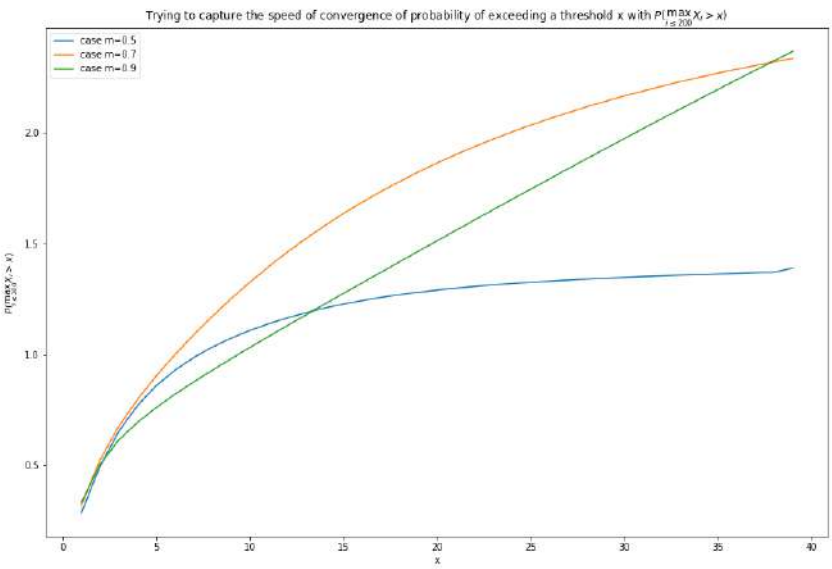

Figure 11: $P(\max_{i\in\mathbb{N}}(X_i) > x)m^{-x}x$ for different $m < 1$
Speed of convergence for $m < 1$

2.3 Galton-Watson process with two environments

In this section, we are given two environments occurring each at a given probability. Let 1 be the favorable environment and 0 the unfavorable one. The populations reproduce in each environment following the same law as at the preceding section of mean m1 for the favorable environment and m0 for the other one. Therefore, m1>m0.

The Kersting lemma generalizes to $\frac{X_n}{\prod_{j<n} m_{i_j}}$ converge almost surely, where $(i_j)_j$ corresponds to the sequence of environments encounter by the trajectory. This is illustrated by the following figure.

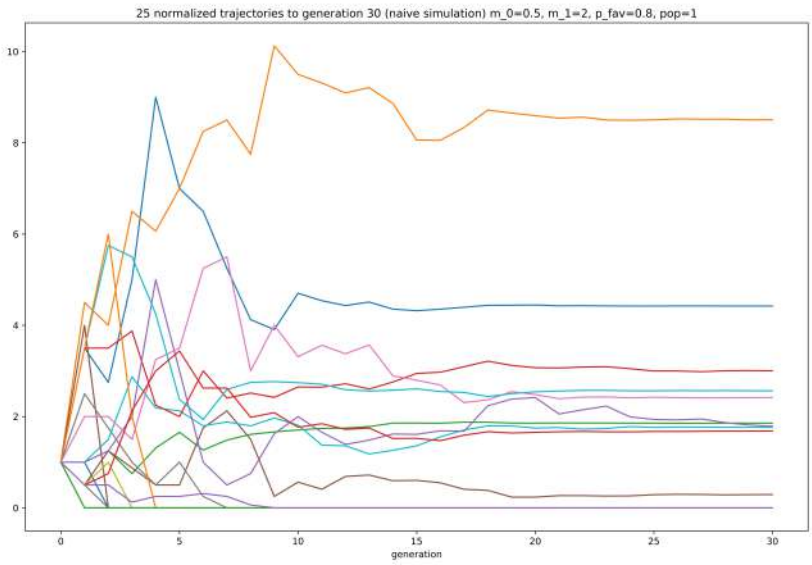

Figure 12: 25 normalized trajectories
Problem parameters : $m_0 = 0.5$, $m_1 = 2$, $p_{fav} = 0.8$, $n = 30$, $pop = 1$

Regarding the use of the Importance sampling in the case of two environments, few changes can be made to adapt the method. Now our Markov chain is indexed by the (m0,m1,pfav) parameters, we can then do a probability change to any other Markov chain indexed by (m'0,m'1,pfav'). The transition kernel of the two environment Markov Matrix can be written as such:

$$P(i,j) = \text{pfav}P_1(i,j) + (1 - \text{pfav})P_0(i,j)$$

with P_1 and P_0 the transition matrices of respectively the geometric GW process of mean 1 and 0

This formula can easily be computed through the negative binomial probability function, however it has its limits for largely different environments $m0 \to 0$ and $m1 \to \infty$, since always one of the terms is very small and can cause an overflow.

Unlike the previous part, the **annealed** survival probability (mean probability over all sequences of environments) can only be computed through a sum on the 2^n sequences of environments.

$$\mathbb{P}(X_n = 0) = \sum_{(i_0,\dots,i_{n-1})\in 0,1^n} p_{i_0}\dots p_{i_{n-1}}(1 - \frac{1}{1 + \frac{1}{m_{i_0}} + \dots + \frac{1}{m_{i_0}\dots m_{i_{n-2}}} + \frac{1}{m_{i_0}\dots m_{i_{n-1}}}})$$

Such a formula has the advantage of being deterministic, however we can not exceed a maximal number of generations because of its exponential time complexity.

The figure 13 shows how a naive Monte Carlo Method gives the same result as the deterministic one. The Monte Carlo has the benefit of being linear on both the maximum number of generations and the number of samples drawn. Consequently, as long as we stay under $\simeq 13$ generations, the deterministic method is better, whereas, the Monte Carlo is a good fit for further generations.

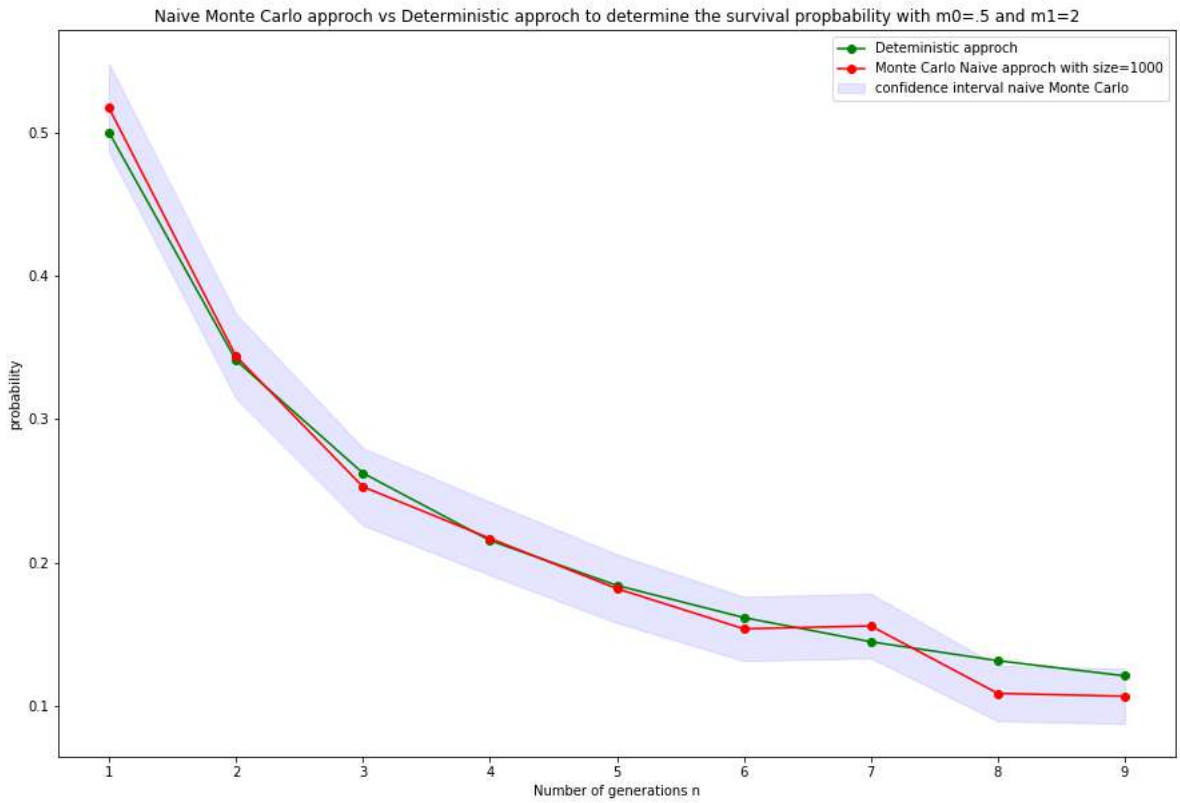

Figure 13: Estimating the probability of survival for the process with two environments
Probability of survival

The figure 14 illustrate well the utility of the importance sampling method when the event is rare enough for the naive Monte Carlo method not to detect it.

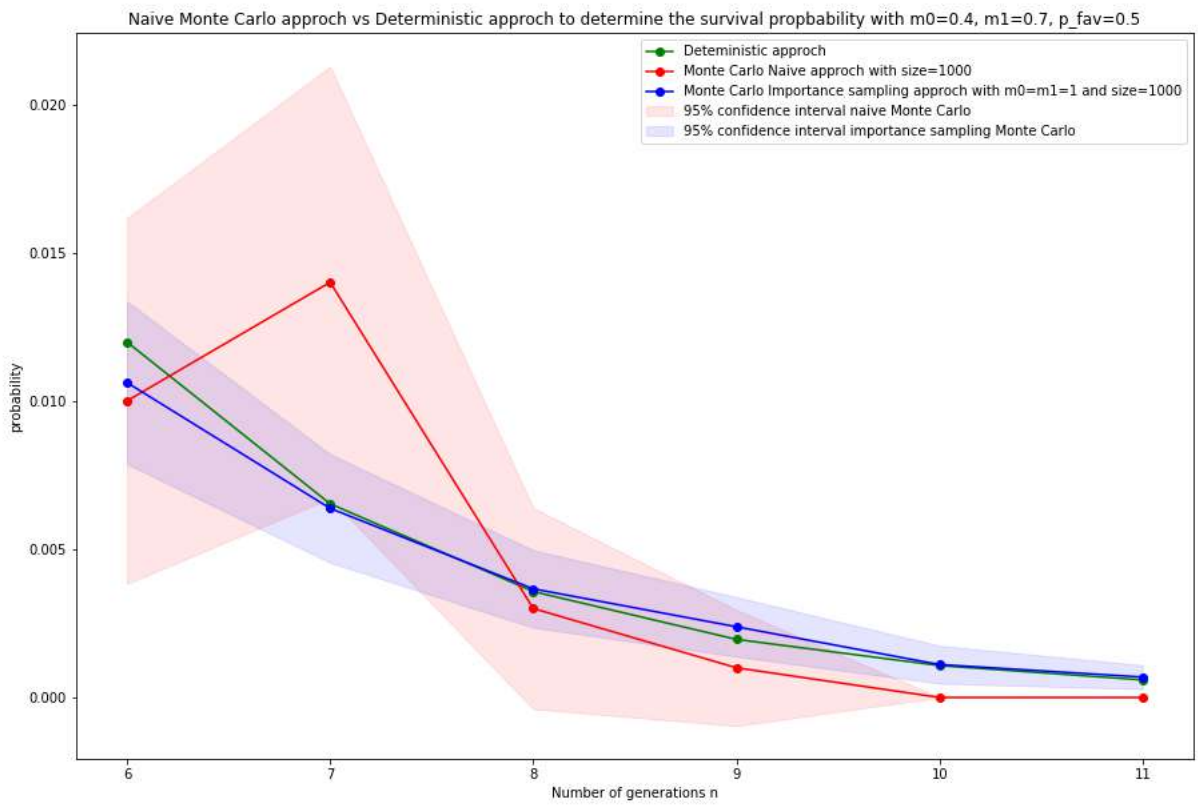

Figure 14: Estimating the probability of survival for the process with two environments on a rarer event

Probability of survival

As stated and proved several times ([JL03]), the critical case for the multi-environment G.W model is when

$$\mathbb{E}_{p_{fav}}(\log m) = p_{fav}\log(m_1) + (1 - p_{fav})\log(m_0) = 0$$

Another interesting regarding the speed of convergence of probability of survival is proved in [JL03]. In the critical case we have under verified hypotheses on finite sums:

$$\mathbb{P}(X_n > 0) \sim \frac{\beta}{\sqrt{n}}$$

Let us first notice that adding a variance on the state makes the convergence to 0 slower in the critical case. We could have safely assumed at this point that β is an increasing function of the variance of the distribution $p_{fav}(1 - p_{fav})(m_1 - m_0)$.

We will try in this part to quantify the effect of the variance of our distribution on that β coefficient

To stay in the critical regime, we fixed $p_{fav} = 1/2$ and iterated over different values of m_1 and $m_0 = \frac{1}{m_1}$

Using only the graph 15 seems to comfort our conjecture about the effect the variance has on β.

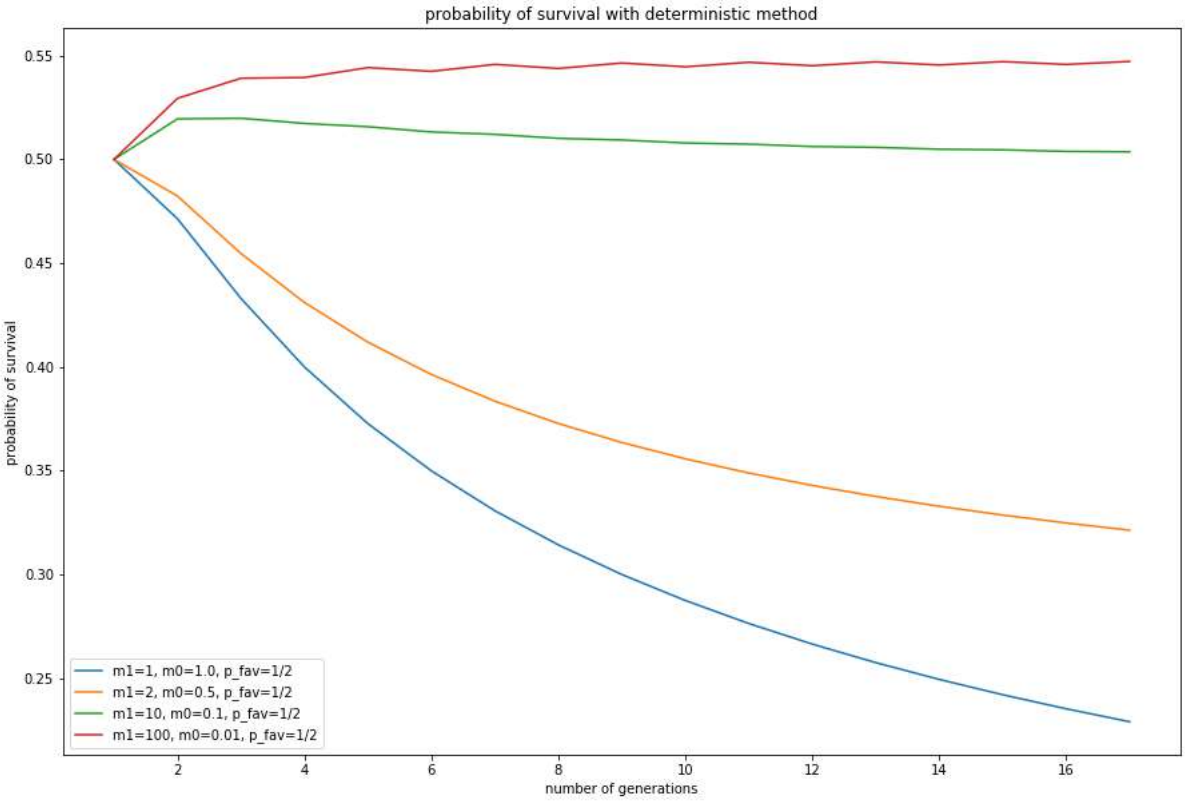

Figure 15: Plotting $\mathbb{P}(X_n > 0)\sqrt{n}$ to verify the result
Probability of survival

Let us see what happens when we increase the number of generations and compute the probabilities with a monte Carlo method.

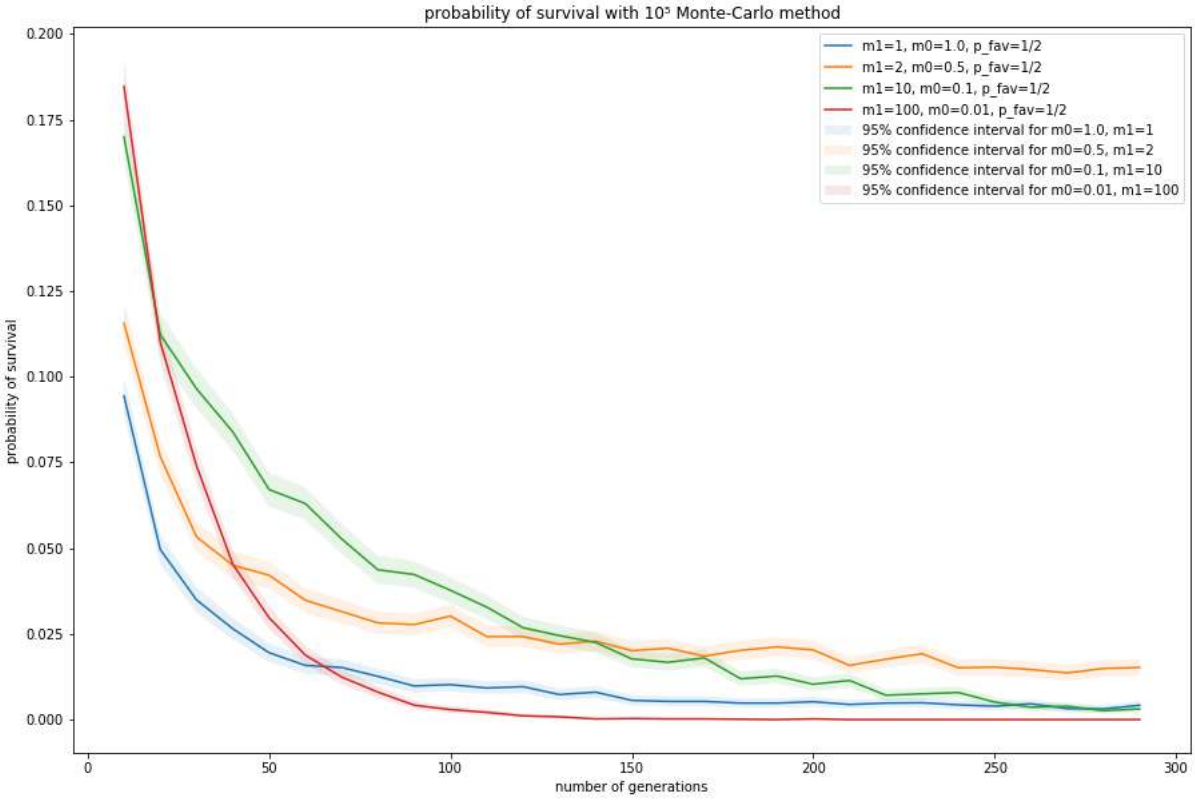

Figure 16: Plotting $\mathbb{P}(X_n > 0)$ to verify the result
Probability of survival in further generations

Unlike what we would have expected the red curve ($m0 = 1/10, m1 = 1$) gets under the remaining ones as the number of generations increases (graph 16). Plotting different values of $\beta = \lim \mathbb{P}(X_n > 0)\sqrt{n}$ as a function of the variance of the critical state we are in, gives the results in the graph 17. Against, what we might have expected, the variance has a different effect on the speed of convergence and the slowest speed of convergence seems to be for the case with $m_0 = 0.5$ and $m1 = 2$.

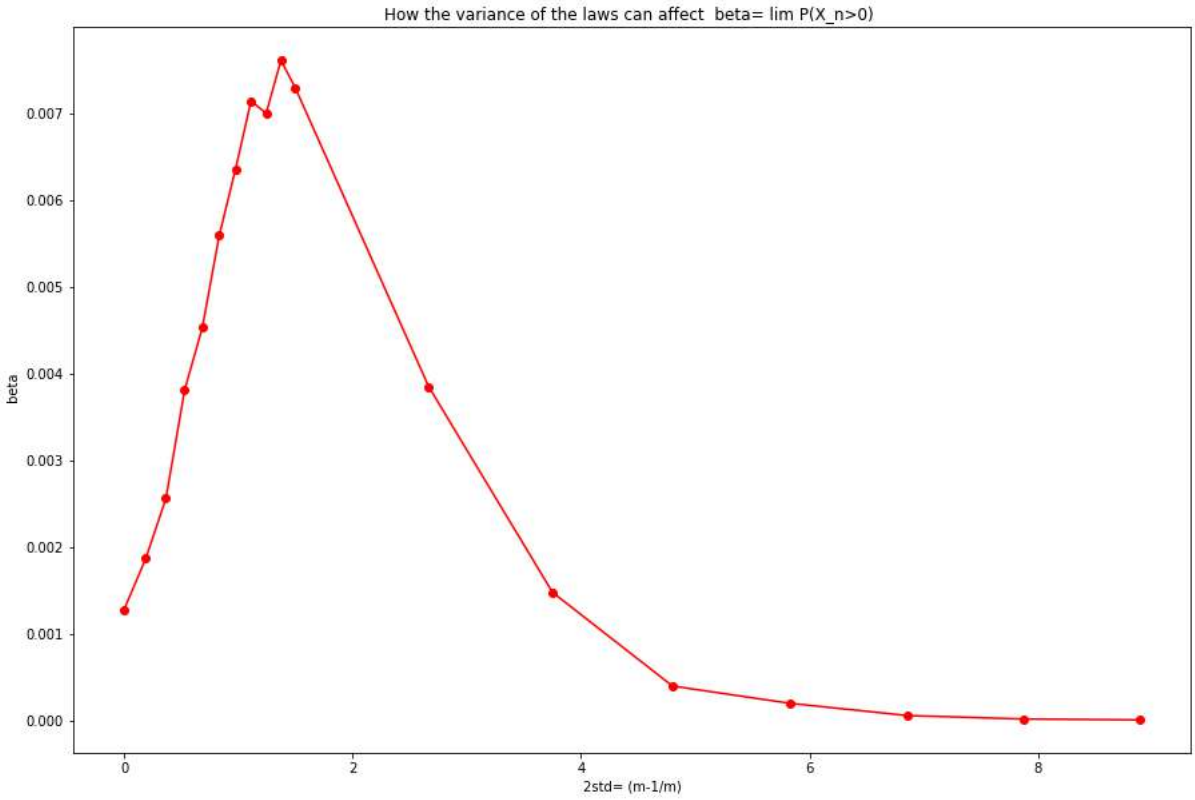

Figure 17: Plotting $\lim \mathbb{P}(X_n > 0)\sqrt{n}$ for different std deviations to verify the result

At last some of the following figures seem to support the theoretical results on the speed of convergence under different cases of the subcritical regime shown in [JL03], event though the Monte-Carlo method gets noisy at the end.

For the case of a strong subcritical regime, very few things can be said on how the distribution may affect $c_1 = \lim \mathbb{P}(X_n > 0)m^{-n}$ since we now have the intervention of the quantity $p_{fav}\log(m_1)m_1 + (1-p_{fav})\log(m_0)m_0$ that may have a stronger effect on the speed of convergence. Note that a general formula for c_1 was given in [PE 72] equation **3.7** with no evident simplification in the case of fractional generating function, (our case).

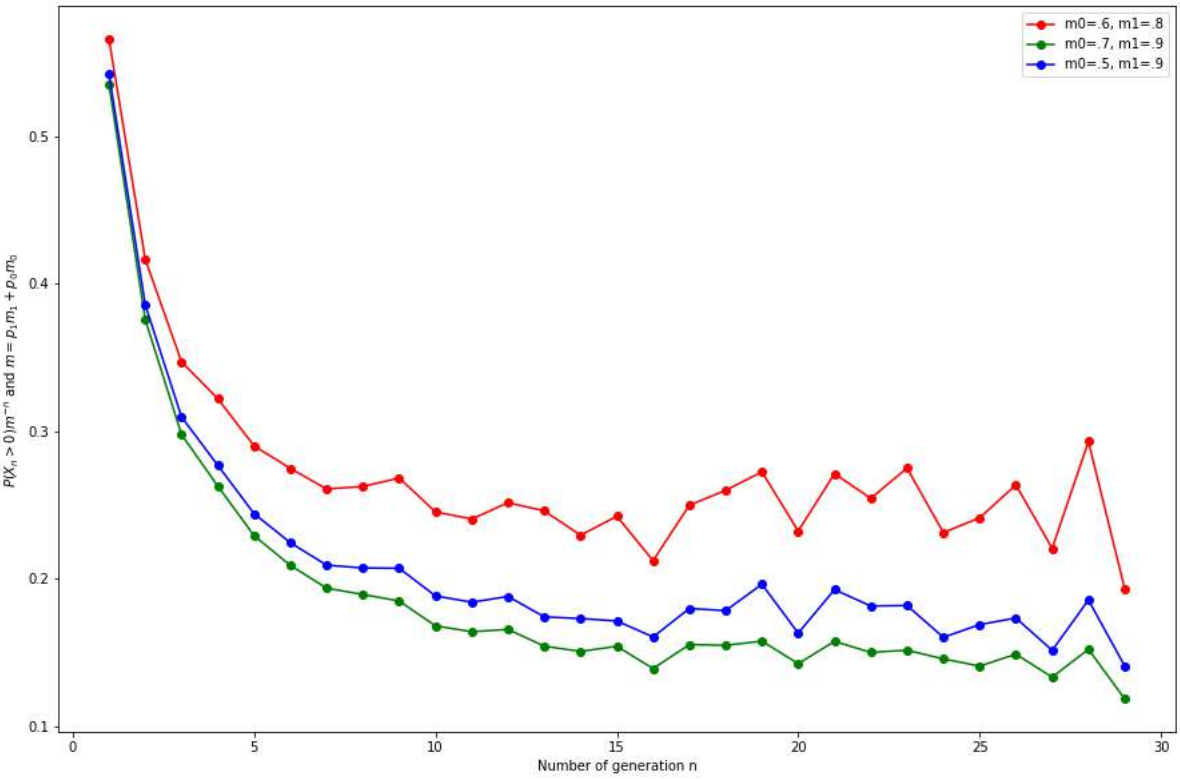

Figure 18: Strong subcritical case example of convergence in m^n

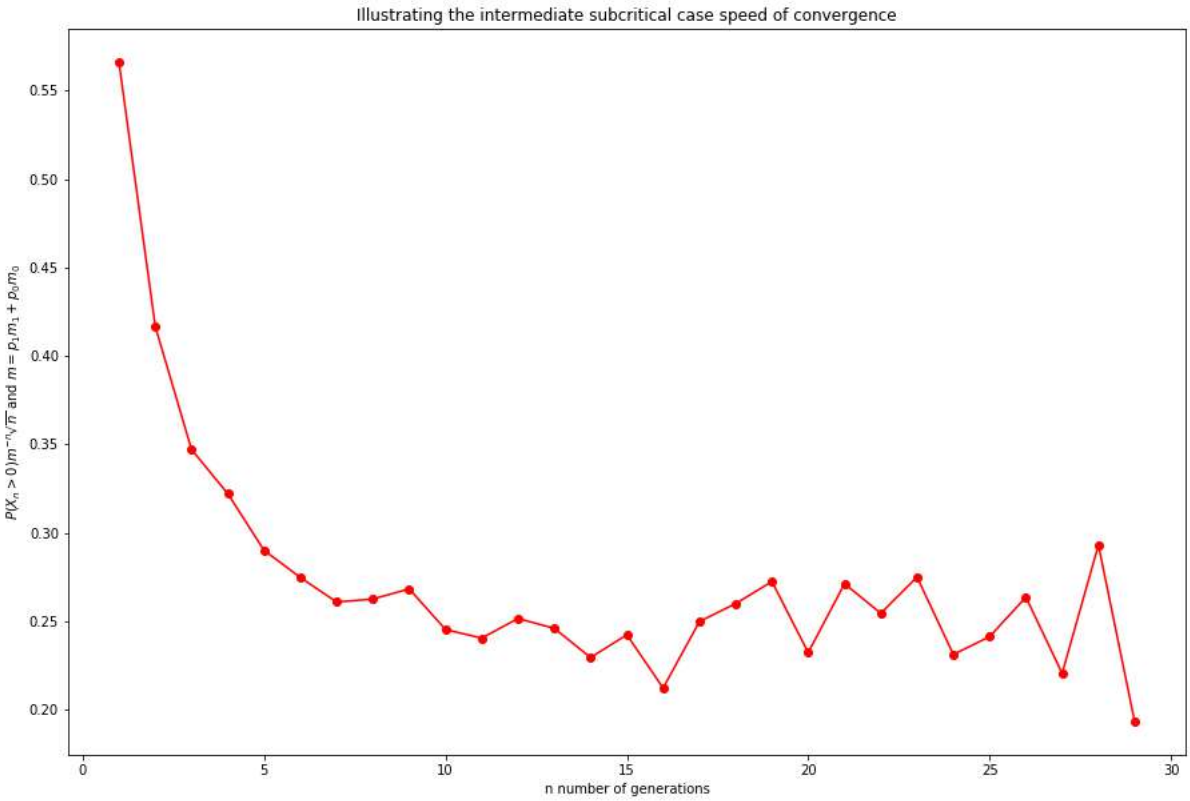

Figure 19: Intermediate subcritical convergence in $m^n/\sqrt{n}$

3
INTERACTING PARTICLE SYSTEMS

3.1 Motivation and presentation

Our project deals with quantities that can be expressed as the expected value of a function over a trajectory of a Markov chain (we call a trajectory of a random process $(X_n)_{n\in\mathbb{N}}$ a realization of the random variable $X_{0:n} := (X_0, ..., X_n)$ given an integer n). For example, $\mathbb{P}(X_n > 0) = \mathbb{E}(\mathbb{1}_{X_n>0})$ where $\mathbb{1}_{X_n>0}$ if a function over $X_{0:n}$. It was therefore very natural at least to try to implement interacting particle systems techniques not only to estimate those quantities, but also to get typical scenarii.

We give here a very brief explanation of how interacting particle systems work from an algorithmic perspective. For theoretical explanations of this, in particular why it works under reasonable conditions see [PDe05].

To build an estimator Suppose you want to estimate the quantity $\mathbb{E}_{\mathbb{P}}(f(X_0, ..., X_t))$ where $(X_t)_{t\geq 0}$ is a Markov chain, $\mathbb{P}$ describes the transition probabilities, and f is a function.

First, define a family of non-negative potentials $G_p(X_{0:p})$ for every integer p, and a simulation parameter M. The choice of the family of potentials is crucial to get a good estimator, we will say a few words about this later.

Start with M "particles" which are a set of trajectories all equal to a fixed initial value X_0. We will build a set of M trajectories $(X_{0:p}^{p,m})_{1\leq m\leq M}$ of length $p+1$ for each integer $0 \leq p \leq t$ (the first upper indice is here to precise that those are the trajectories obtained at step p).

For each integer $0 < p \leq t$, perform successively the two following processes :

1. selection. Choose M i.i.d. indices $I_1, ..., I_M$ in $\{1, ..., M\}$ with the following probability : $\mathbb{P}(I_j = m) \propto G_{p-1}(X_{0:p-1}^{p-1,m})$

2. mutation. $X_{0:p}^{p,m}$ is obtained by adding a last value $X_p^{p,m}$ to $X_{0:p-1}^{p-1,I_m}$ randomly under the probability $\mathbb{P}(X_{p-1}^{p-1,I_m}, .)$

At each step until $t-1$, calculate the constant C_p defined by $C_0 = 1$ and $C_p = C_{p-1}\frac{1}{M}\sum_{m=1}^{M} G_p(X_{0:p}^{p,m})$.

Finally, we have the following estimation : $\mathbb{E}_{\mathbb{P}}(f(X_0, ..., X_t)) \approx C_{t-1}\frac{1}{M}\sum_{m=1}^{M}\frac{f(X_{0:t}^{t,m})}{\prod_{p=1}^{t-1} G_p(X_{0:p}^{t,m})}$.

To get typical scenarii Picking trajectories among $(X_{0:t}^{t,m})_m$ with a probability proportional to $\frac{1}{\prod_{p=1}^{t-1} G_p(X_{0:p}^{t,m})}$ for each m is a generator of trajectories under the law described by $\mathbb{P}$. Therefore, to simulate trajectories conditional to an event A, we just need to :

1. remove all the trajectories that don't satisfy the condition A

2. pick trajectories among those which remain with a probability given by the formula below

About the simulation parameter M We need to choose a value of the simulation parameter M to run our algorithm that estimates the quantity we are looking at. A clear drawback of a big value of M is the runtime, which is linear in M. It is in fact the only drawback. An advantage of a big value of M is a smaller variance of the estimator, plus the fact that the consistency is not granted for really small values of M. In short, it is a variance-runtime trade-off.

For typical scnenarii, a small value of M can lead to an imbreeding situation. We will show such situations later.

About the choice of potentials The basic idea is to choose a family of potentials that will favor "interesting" trajectories (in the light of the event we want to estimate the probability) in the selection process. For example, if you want to estimate the probability $\mathbb{P}(X_n > c)$ for a very big value of c, you will choose potentials that favor "high" trajectories. By doing so, you transform a rare event into a non-rare one, and you get a estimator that surpass a naive Monte-Carlo estimator both in variance and runtime. In fact, for very rare event, a naive Monte-Carlo estimator just doesn't work with a reasonable runtime (you will get 0 for an estimation of the probability you're looking at).

We will discuss in depth the choice of good potentials in specific situations later.

3.2 Code's architecture

We wrote a code in a modular way in the Jupyter notebook `Interacting Particle Systems.ipynb`. One advantage of this is to isolate mistakes and therefore to detect and correct them quickly. More importantly, it is very convenient in the exploratory phase of our project. Indeed, if we want to test a new potential, we just have to define it in a new cell and then to use it as an argument of the right function.

Through all this notebook, we use the following notations :

- m is the expected value of the number of children of an individual
- n denotes the generation
- pop is the size of the initial population (at generation 0)
- M is a simulation parameter : it is the number of interacting particles in our simulation.

For the second part of the project, we also use :

- p_{fav} the probability to select a favorable environment at a given generation
- m_1 is the expected value of the number of children of an individual in a favorable environment
- m_0 is the expected value of the number of children of an individual in a unfavorable environment

The two main functions of this Jupyter notebook are :

1. `trajectories(potential)` : simulate, with the given potential, M trajectories of length n with the selection-mutation process, along with the normalizing constant C. The reason for this is that we can't calculate this constant with the final trajectories, since some trajectories have been erased during the selection process. M and n are global parameters and are therefore not arguments of this function.

2. `proba(event, potential)` : estimate the probability of the event `event` with trajectories simulated with the potential `potential`. Of course, `event` and `potential` have to be defined in a cell before. Function `proba` uses function `trajectories` to do its estimation, using the formula $\mathbb{E}_{\mathbb{P}}(f(X_0, ..., X_t)) \approx C \frac{1}{M} \sum_{m=1}^{M} \frac{f(X_{0:t}^{t,m})}{\prod_{p=1}^{t-1} G_p(X_{0:p}^{t,m})}$

In the second part of the project, we still deal with a Markov chain so the interacting particle systems approach is still supposed to work theoretically. We wrote a function `trajectories2` that also returns the sequence of environments (favorable or unfavorable) encounter by each trajectory because we will study it, as well as a function `proba2`.

3.3 Confidence intervals

We want to associate a confidence interval to an estimated probability. That is the reason why we defined the function `confidence(X, gamma)` wich returns a confidence interval of confidence level gamma given an array X of independant realizations of an estimator.

We use the following formula :

$$[\frac{1}{L}\sum_{l=1}^{L} p_l - Stud^{-1}(\frac{1+\gamma}{2}, L-1)\frac{\sigma_L}{\sqrt{L}}, \frac{1}{L}\sum_{l=1}^{L} p_l + Stud^{-1}(\frac{1+\gamma}{2}, L-1)\frac{\sigma_L}{\sqrt{L}}]$$

where σ_L is the empirical standard deviation of the realizations $(p_l)_{l=1,...,L}$

We use the function `confidence` many times in the notebook.

3.4 Around survival in under-critic case

Under-critic case is defined by $m < 1$. The event $X_n > 0$ becomes rare when n grows : its probability tends to zero. We interest ourselves to this event in this subsection.

3.4.1 Choosing the best potential

In interacting particule systems, the key is to choose the right potential in function of what you want to estimate in order the get best possible estimator. In fact, it is the only thing we choose, except the simulation parameter M.

We will present here a heuristic to guess what a good potential might be, in our specific example. A good potential must :

- "turn the rare event into a non-rare one". In order to have a good estimator of the probability of survival, the part of the estimator dealing with simulated trajectories must have a not too high variance. This imply that the survival probability over the simulated trajectories $(\frac{1}{M}\sum_{m=1}^{M} \frac{1_{X_n^{n,m}>0}}{\prod_{p=1}^{t-1} G_p(X_{0:p}^{t,m})})$ have to be not too small. So the potential must select "high trajectories".
- be not too selective. If the potential favors the high trajectories too much, it will over-select them in the selection process, leading to a very high imbreeding of the simulated trajectories. This is also very bad for the variance of the estimator, the trajectories being very correlated.

Therefore, a natural choice of a good potential would be `survival potential` defined as 1 if the last value of the trajectory is positive, and 0 if it is 0. Indeed, it is good at making the survival probability of the trajectories not too small because they can die only at the very last generation in the mutation process. And it is very good at not being too selective because it doesn't discriminate at all between the alive trajectories.

We thought to other potentials. Here is the list of the potentials we will compare :

- `survival potential` : $G_p(X_{0:p}) = 1_{X_p>0}$
- `exponentially increasing potential` : $G_p(X_{0:p}) = e^{\lambda(X_p - X_{p-1})}$
- `exponentially high potential` : $G_p(X_{0:p}) = e^{\lambda X_p}$
- `linearly high potential` : $G_p = X_p$ (remark : there is no sense to parametrize this potential since only the relative values of the potentials of the trajectories matter)

Those potentials might be good in light of the heuristic presented above. Some are potentials with a parameter. For those, we have to find first a nearly optimal value of this parameter. In order to do so, we drew boxplots of the resulting estimators with different value of the parameter, in order to find one value with a low associated variance. Since `exponentially increasing potential` was in fact a bad potential, we present here the approach only for `exponentially high potential`.

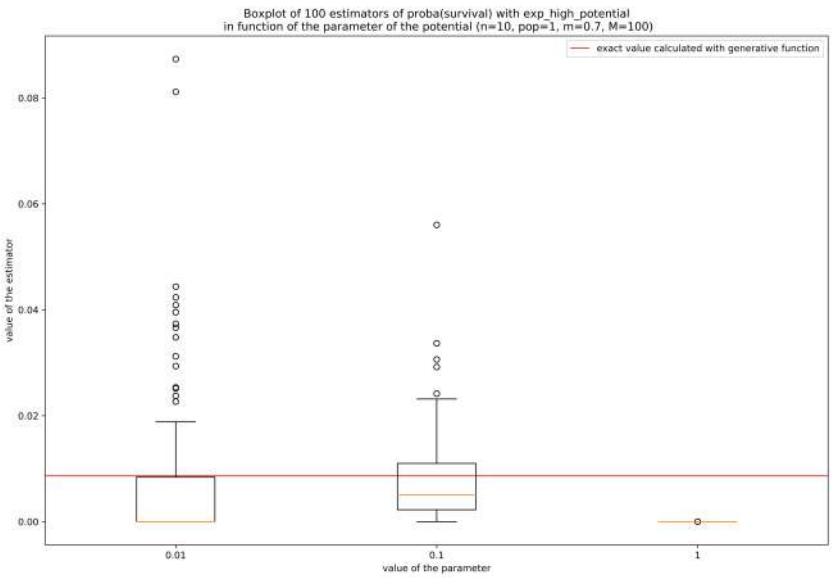

Figure 20: Boxplot of 100 estimators of probability of survival with `exponentially high potential` The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$, the simulation parameter is $M = 100$, and the tested potential parameters are $0.01, 0.1, 1$

Value of parameter λ	0.01	0.1	1
Noise-to-signal ratio	1.91	1.08	6.88

Table 1: Noise-to-signal ratio with `exponentially high potential` in function of parameter λ
The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$, the simulation parameter is $M = 100$. Noise-to-signal ratio = standard deviation divided by mean. This table has been made with the same realizations of the estimator as the figure above.

For `exponentially high potential`, 0.1 appears to be the best value among those on the boxplot (for the given values of the problem parameters). 1 is obviously a very bad value because the estimator always return 0 which gives no information. 0.01 is almost as good as 0.1 in term of standard deviation, but returns 0 at least half of the time.

A remark on the boxplot : it seems that the median of the estimators are smaller than the exact estimated value. It is not a problem as long as the expected value of the estimator is correct, since in practice we will make many independant simulations to estimate the desired values.

Then we compare the potentials between them to find the best.

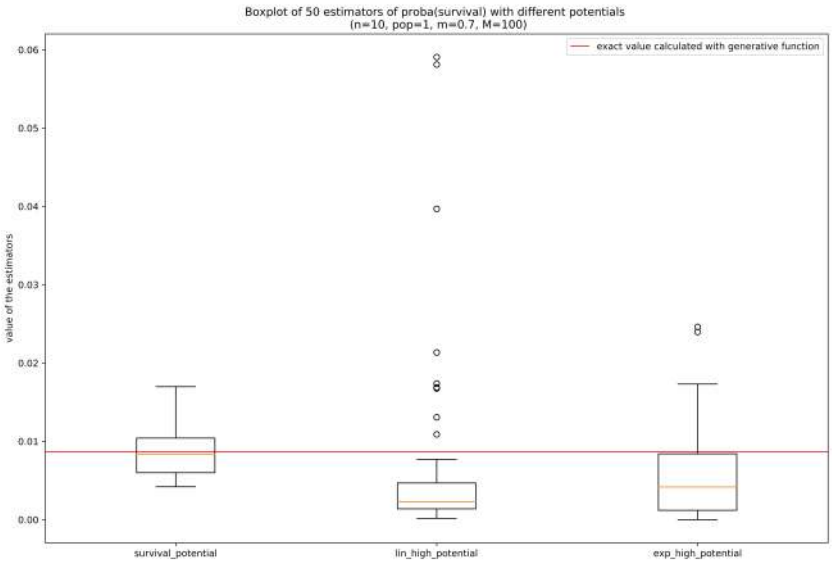

Figure 21: Boxplot of 50 estimators of probability of survival with three different potentials
The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$, the simulation paramter is $M = 100$. For `exponentially high potential` $\lambda = 0.1$

Potentials	`survival`	`linear`	`exponential`
Noise-to-signal ratio	0.36	1.81	0.98

Table 2: Noise-to-signal ratio three different potentials
The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$, the simulation parameter is $M = 100$. Noise-to-signal ratio = standard deviation divided by mean. This table has been made with the same realizations of the estimator as the figure above.

`survival potential` is the best : it has by far the smallest variance.
Two important remarks :

- estimators are random variables, thus the boxplots presented here are only one realization of a multidimensional random variable. In practice, we have simulated those boxplots many times and the general outcomes were always the same.
- we have studies it only for fixed values of the parameters. In practice, we have followed the same approach for different values of the parameters and the conclusion was always the same.

We quickly checked the consistency of this estimator, using the law of large numbers. Good news : it is consistent !

Finally, we can study the influence of the simulation parameter M on the estimator. Warning : for a value of M too small, it can crash (if all the trajectories die in the meantime at one step, we can't do the selection process since the value of the potential of each trajectory is 0).

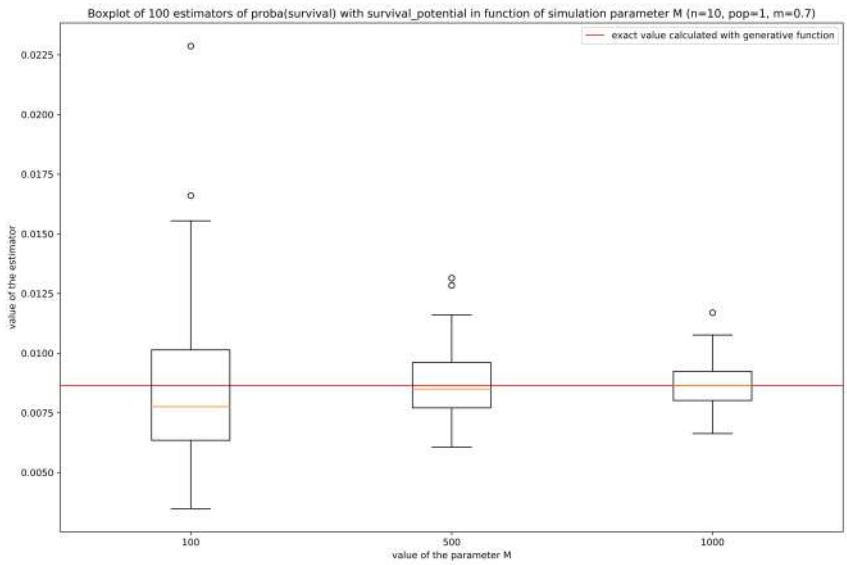

Figure 22: Boxplot of 100 estimators of probability of survival with potential `survival potential` for different values of simulation parameter M

The tested values of simulation parameter M are $100, 500, 1000$. The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$

Values of M	100	500	1000
Noise-to-signal ratio	0.38	0.16	0.11

Table 3: Noise-to-signal ratio with `survival potential` for different values of simulation parameter M

The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$. Noise-to-signal ratio = standard deviation divided by mean. This table has been made with the same realizations of the estimator as the figure above.

Conclusion : `survival potential` is the best potential we have found for events around the survival in the sub-critical case. The only thing we can set is the simulation parameter M, since `survival potential` is a parameter-free potential. We face a variance versus runtime trade-off. In practice, $M = 100$ works well. We can often go up to $M = 1000$ because the runtime are not that long (but of course if also depends on n...)

3.4.2 Study of probability of survival

We have two theoretical results concerning the probability of survival, i being here the size of the initial population.

$$\frac{i(1-m)m^n}{1+m-2m^{n+1}} \leq \mathbb{P}(X_n > 0) \leq m^n i$$

$$\mathbb{P}(X_n > 0) = 1 - (\frac{1-m^{n-1}}{1-m^{n+1}})^i$$

We will use the second theoretical result as a guarantee that our estimator is correct, and we will illustrate the first one with our simulations. In order to do so, we plot on the same figure a confidence interval at level 0.95 obtained with our estimator, the theoretical value, and the theoretical bounds. Here are figures we obtained.

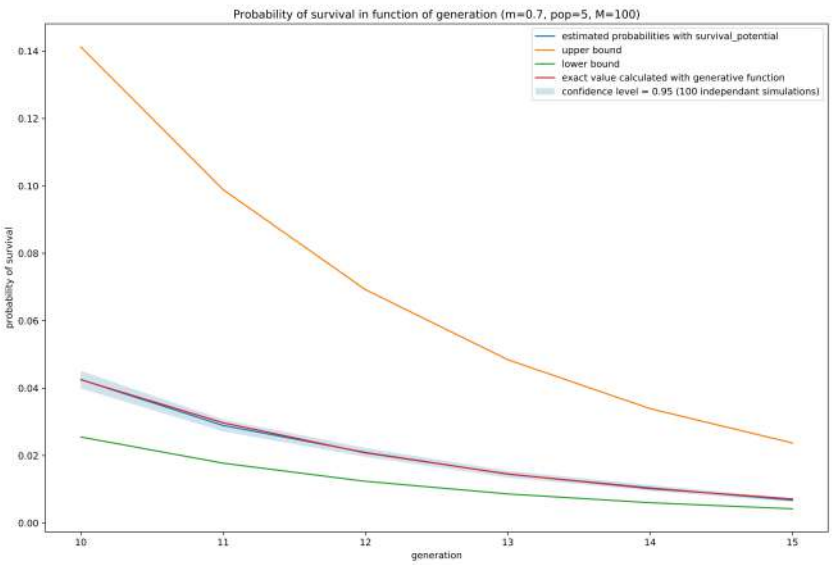

Figure 23: Probability of survival from generation 10 to 15
The problem parameters are $pop = 5$, $m = 0.7$, simulation parameter $M = 100$.

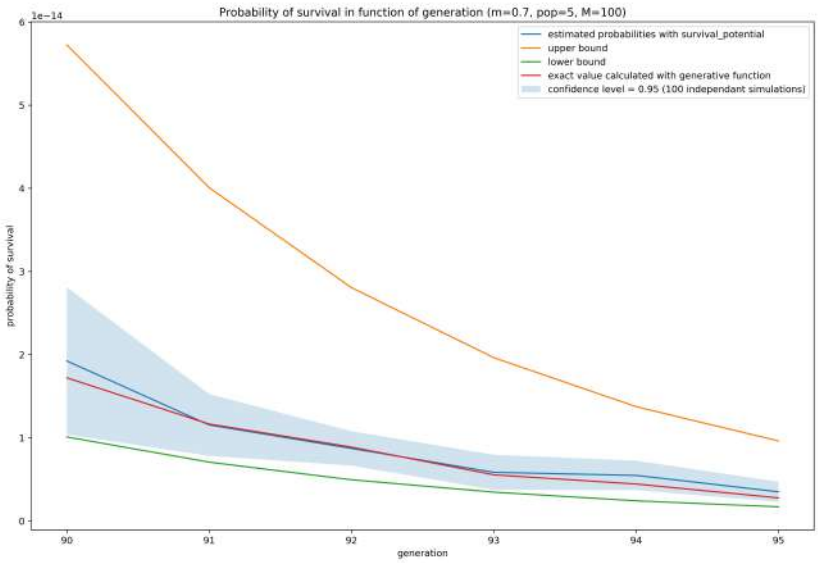

Figure 24: Probability of survival from generation 90 to 95
The problem parameters are $pop = 5$, $m = 0.7$, simulation parameter $M = 100$.

From those four figures, we conclude that :

- our estimator works well, no matter the values of parameters n, pop
- it works even to estimate probabilities around 10^{-15}
- nevertheless, the noise-to-signal ratio is bigger when the probability is smaller (relative width of the blue area)
- everything is coherent with the theoretical results

To see the influence of parameter m We have seen what the curve look like for only one value of parameter m. Here we simulate it with two different values of this parameter to see the impact it has on the probability of survival.

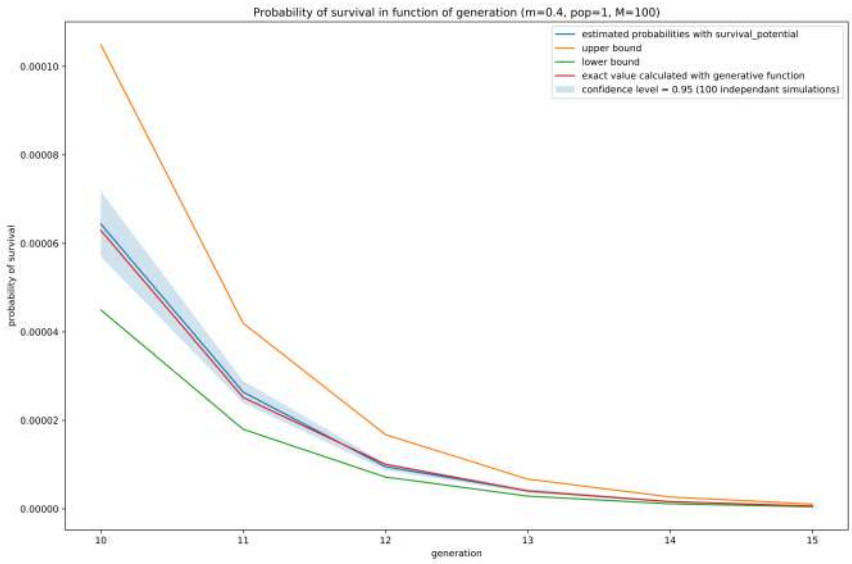

Figure 25: Probability of survival from generation 10 to 15
The problem parameters are $pop = 1$, $m = 0.4$, simulation parameter $M = 100$.

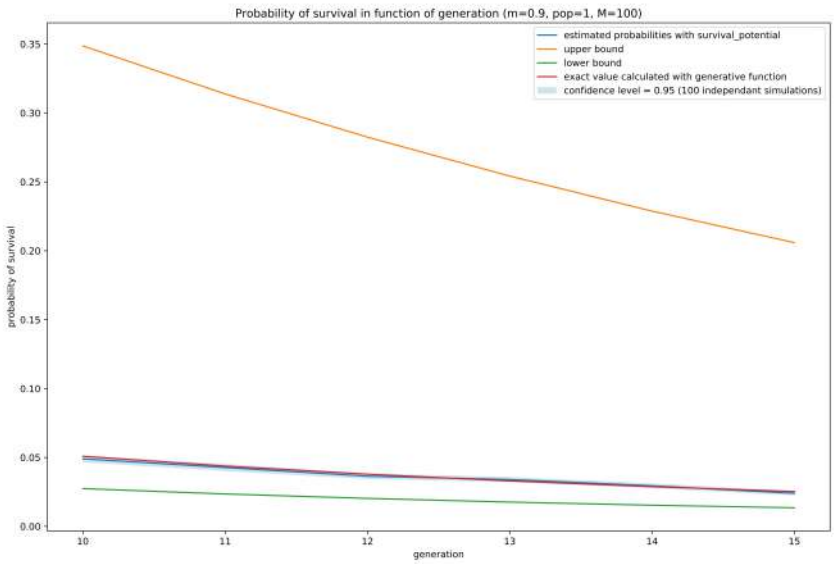

Figure 26: Probability of survival from generation 10 to 15
The problem parameters are $pop = 1$, $m = 0.9$, simulation parameter $M = 100$.

From those two figures, and from the four above, we can conjecture that :

- the probability of survival decreases exponentially in n. Indeed, the ratio $\frac{X_{n+5}}{X_n}$ seams to be nearly constant.
- the slop of this decrease is steeper for small values of m
- the probability of survival is decreasing in m

Thoses conjectures are true since $\mathbb{P}(X_n > 0) \sim (1 - m^2)m^{n-1}$ when $pop = 1$.

3.4.3 Distribution conditional to survival

.

In this subsubsection, we are looking at the distribution of X_n conditional to $X_n > 0$. We will use interacting particle systems with `survival potential` since we have shown that it is the most appropriate in this situation. To do so, we had two non-naive approaches :

1. estimate conditional probabilities $\mathbb{P}(X_n = k \mid X_n > 0)$ straight with the formula

$$\mathbb{E}_{\mathbb{P}}(f(X_{0:t}) \mid A) = (\sum_{m=1}^{M} \frac{f(X_{0:t}^{t,m}) 1_{X_{0:t}^{t,m} \in A}}{\prod_{p=1}^{t-1} G_p(X_{0:p}^{t,m})})(\sum_{m=1}^{M} \frac{1_{X_{0:t}^{t,m} \in A}}{\prod_{p=1}^{t-1} G_p(X_{0:p}^{t,m})})^{-1}$$

   With $A = \{X_n > 0\}$ and $f = 1_{X_n = k}$ we can estimate the probabilities $\mathbb{P}(X_n = k \mid X_n > 0)$.

2. pick a large number of typical survival scenarii, and estimate those probabilities by the proportions of the last values of those trajectories. See the next subsubsection for more details.

From an algorithmic perspective, they are almost the same. Indeed, in practice the first approach estimates the probabilities as a proportion on the end of the trajectories. This is because the term $\sum_{m=1}^{M} \frac{1_{X_{0:t}^{t,m} \in A}}{\prod_{p=1}^{t-1} G_p(X_{0:p}^{t,m})}$ is simply equals to the number of trajectories that didn't die at the last generation, because we use `survival potential` as potential.

The first method is logically better. Indeed, in the spirit of the second method, we should pick a small proportion of trajectories among those available to avoid "imbreeding" and to reach the asymptotic independence. There is not this waste of trajectories in the first method, which will probably have a better variance as a consequence.

We implemented both approaches as well as a naive one to be sure that they are correct. Both methods work, and indeed the first one is better (smaller variance for the same simulation parameter M ie. the same runtime). It is therefore the method we will use here.

A theoretical result we will illustrate here (asymptotic conditional distribution) :

$$\mathbb{P}(X_n = j \mid X_n > 0, X_0 = i) \xrightarrow[n \to \infty]{} (1 - m)m^{j-1}$$

Influence of M and K To estimate the conditional probabilities with a confidence interval, we need to run K times the simulation. Of course, increasing K diminish the size of the confidence interval in a clear way (central limit theorem), but increase the runtime. The same is true for parameter M, and the two next figures are here to show the impact of M on the variance of the estimators, which is less clear theoretically.

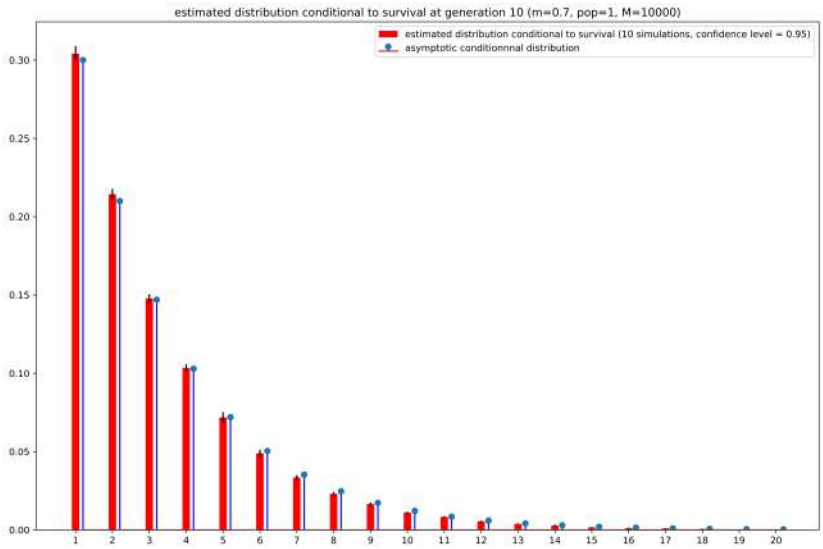

Figure 27: Estimated distribution conditional to survival at generation 10
The problem parameters are $pop = 1$, $m = 0.7$, simulation parameter $M = 10000$, number of simulations $K = 10$

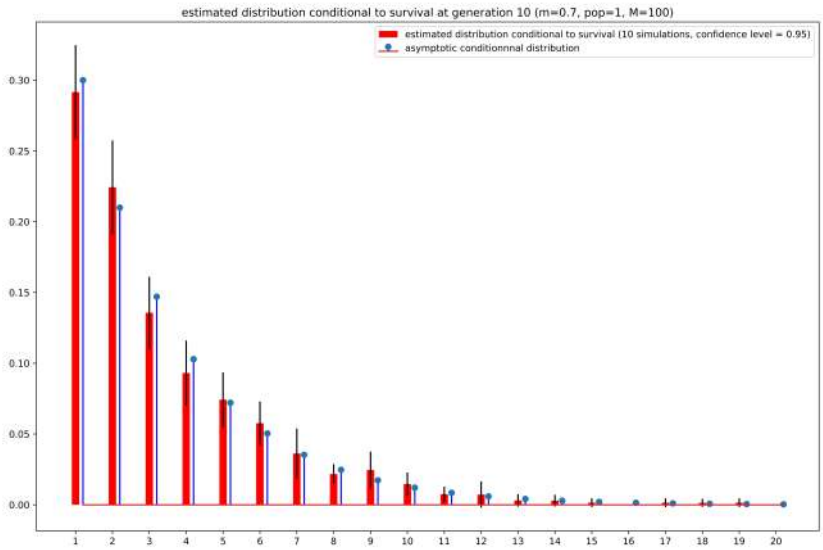

Figure 28: Estimated distribution conditional to survival at generation 10
The problem parameters are $pop = 1$, $m = 0.7$, simulation parameter $M = 100$, number of simulations $K = 10$

Influence of the problem parameters m and pop .

Impact of parameter m. First, m has a influence on the shape of the asymptotic conditional distribution. This is clear when looking at the mathematical expression of it.

Second, the bigger m, the slower the convergence toward the asymptotic conditional distribution. This can be seen on the two following figures.

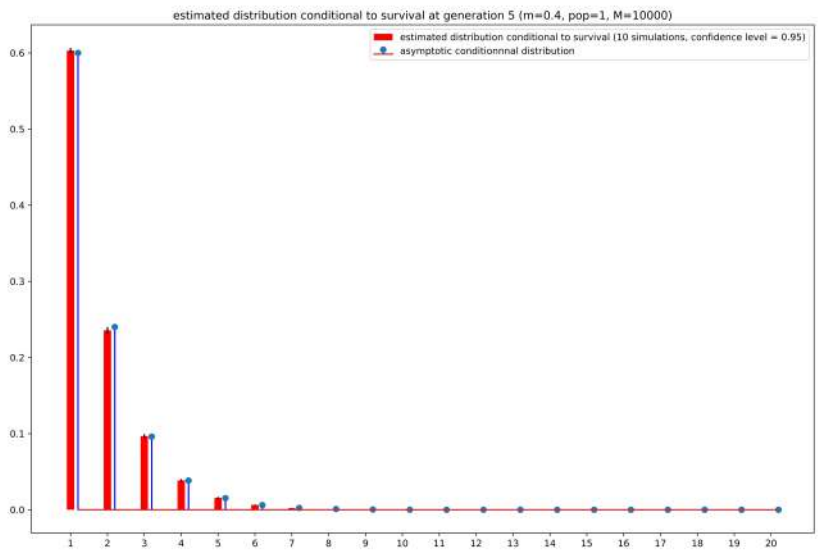

Figure 29: Estimated distribution conditional to survival at generation 5
The problem parameters are $pop = 1$, $m = 0.4$, simulation parameter $M = 10000$, number of simulations $K = 10$

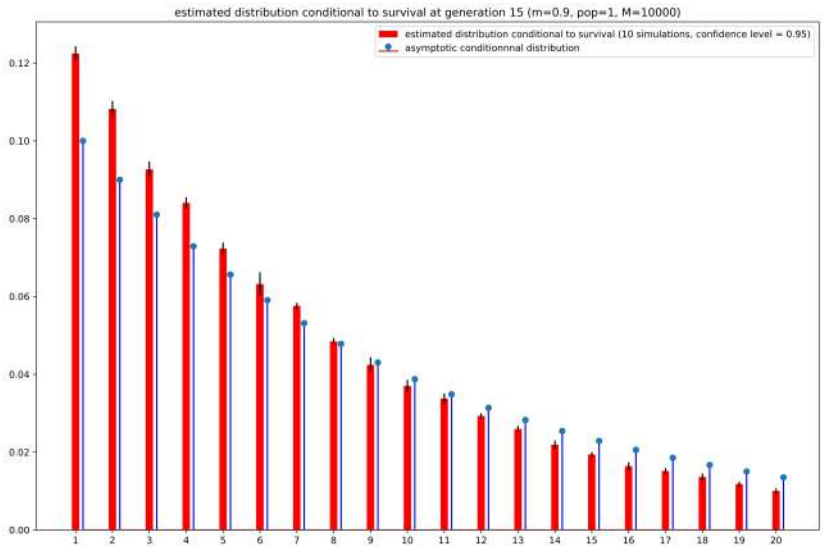

Figure 30: Estimated distribution conditional to survival at generation 15
The problem parameters are $pop = 1$, $m = 0.9$, simulation parameter $M = 10000$, number of simulations $K = 10$

Impact of parameter *pop*. On the contrary to m, *pop* has no impact of the asymptotic conditional distribution. It has a big impact on the shape of the conditional distribution for small values of n ie. before convergence. Indeed :

- for very small values of *pop*, the conditional distribution is steeper than the asymptotic conditional distribution, meaning that the smallest values have a bigger weight and the larger values a smaller weight
- for intermediate values of *pop*, it is exactly the opposite
- for very large values of *pop*, there is even a "wave" shape, with the weights that don't reach their maximum in 1

The limits for "small" and "large" depends on n and m.

The following figures are very convincing. We show figures first for $pop = 1$, then for $pop = 10$ and finally for $pop = 60$

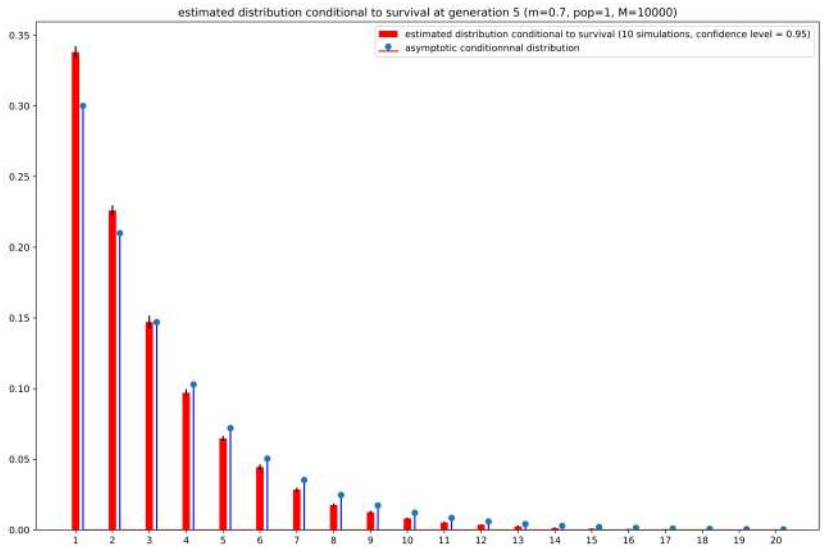

Figure 31: Estimated distribution conditional to survival at generation 5
The problem parameters are $pop = 1$, $m = 0.7$, simulation parameter $M = 10000$, number of simulations $K = 10$

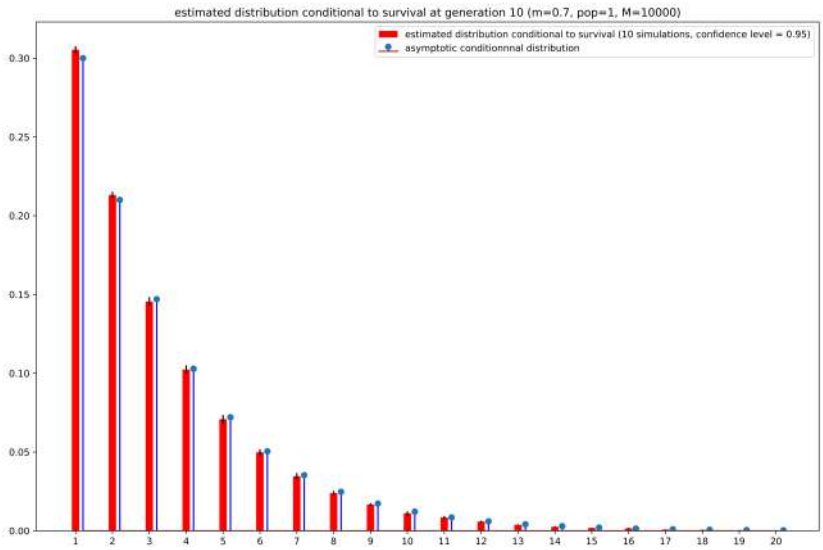

Figure 32: Estimated distribution conditional to survival at generation 10
The problem parameters are $pop = 1$, $m = 0.7$, simulation parameter $M = 10000$, number of simulations $K = 10$

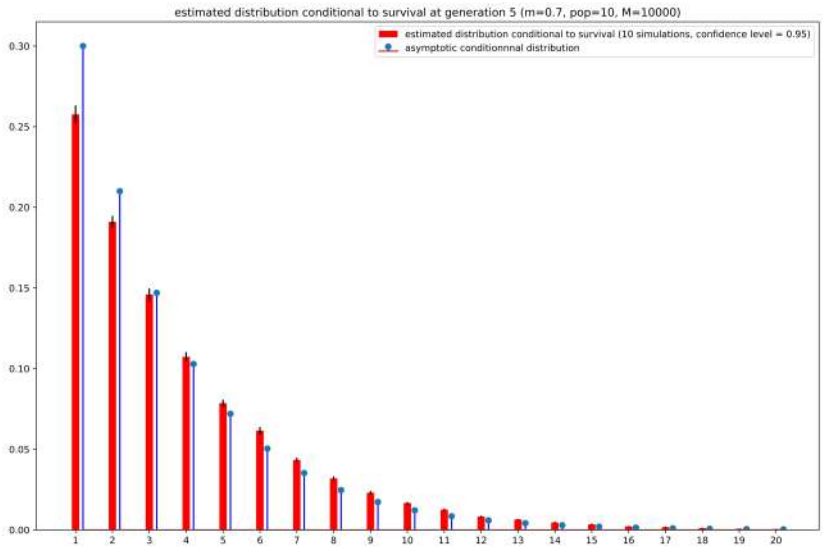

Figure 33: Estimated distribution conditional to survival at generation 5
The problem parameters are $pop = 10$, $m = 0.7$, simulation parameter $M = 10000$, number of simulations $K = 10$

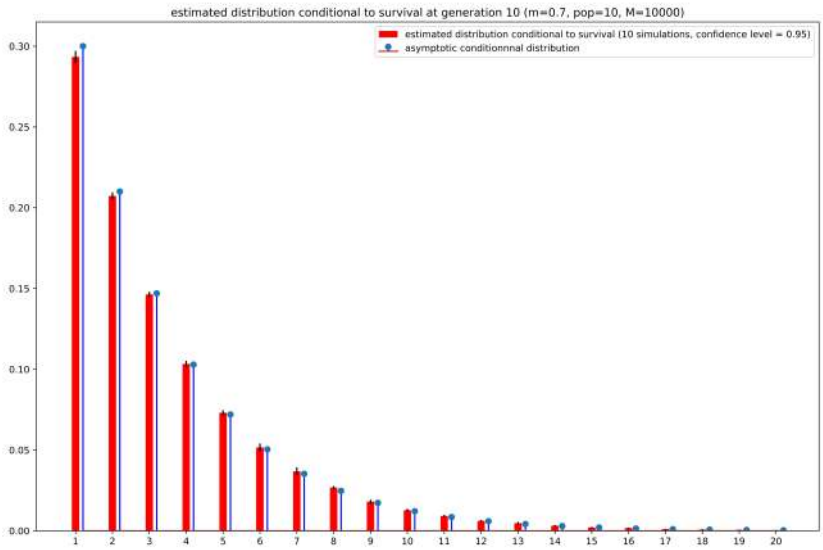

Figure 34: Estimated distribution conditional to survival at generation 10
The problem parameters are $pop = 10$, $m = 0.7$, simulation parameter $M = 10000$, number of simulations $K = 10$

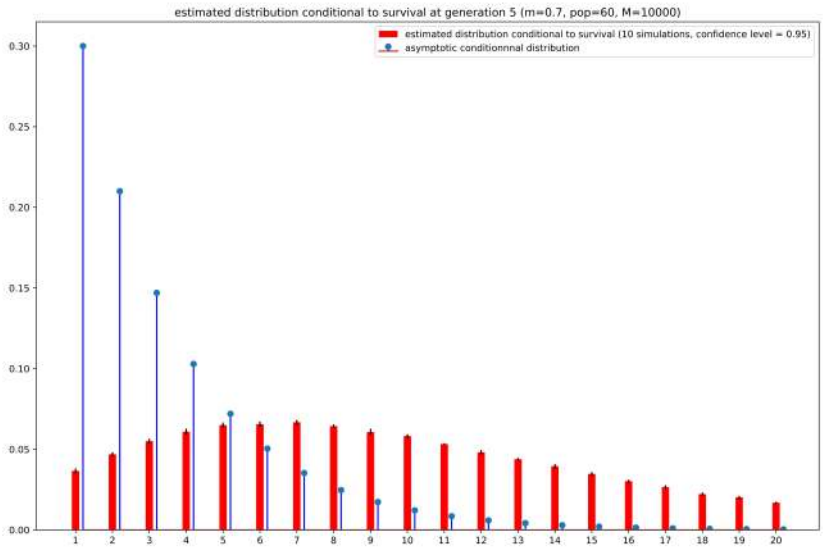

Figure 35: Estimated distribution conditional to survival at generation 5
The problem parameters are $pop = 60$, $m = 0.7$, simulation parameter $M = 10000$, number of simulations $K = 10$

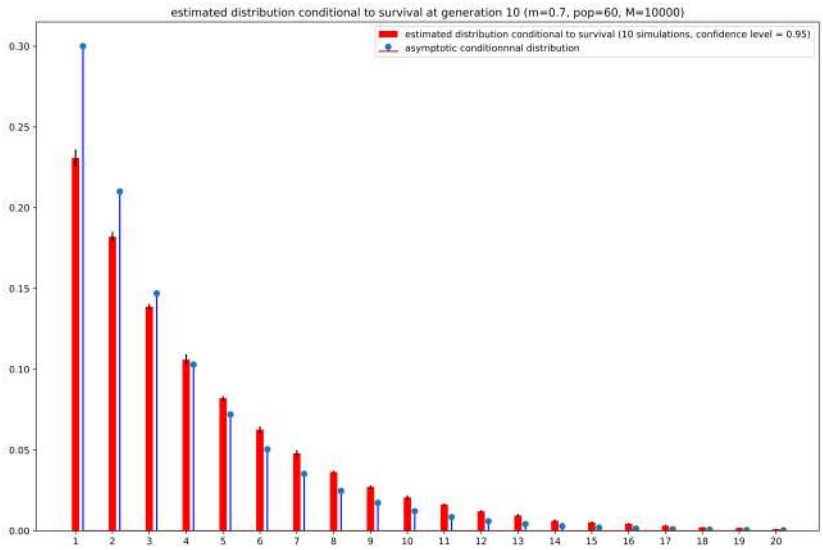

Figure 36: Estimated distribution conditional to survival at generation 10
The problem parameters are $pop = 60$, $m = 0.7$, simulation parameter $M = 10000$, number of simulations $K = 10$

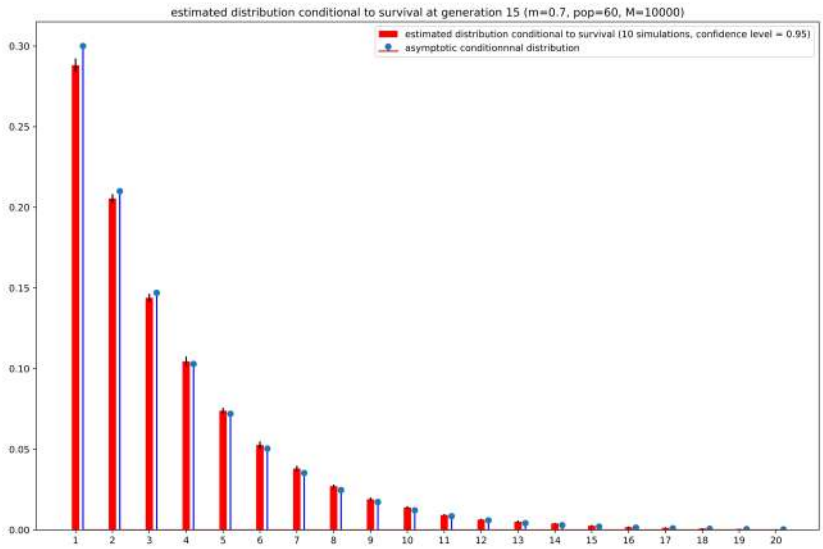

Figure 37: Estimated distribution conditional to survival at generation 15
The problem parameters are $pop = 60$, $m = 0.7$, simulation parameter $M = 10000$, number of simulations $K = 10$

3.4.4 Generation of typical scenarii of survival

.

It is of course interesting to be able to estimate the probability of an event (rare or not), and it is also interesting to know what happens when that event occurs. Here, we try to find typical scenarii that lead to a survival at generation n. Mathematically speaking, we want to simulate trajectories under the law conditional to the survival at generation n.

Here also, interacting particle systems enable us to do it. Using `survival potential`, the weight to apply to trajectories simulated under the selection-mutation process when picking them is uniform on those which are not dead at generation n. See the subsubsection "motivation and presentation" for more details. We just need to pay attention to have a simulation parameter M large enough to avoid an imbreeding situation.

About imbreeding .

The two following figures are here to show the effect of choosing a too small value for simulation parameter M.

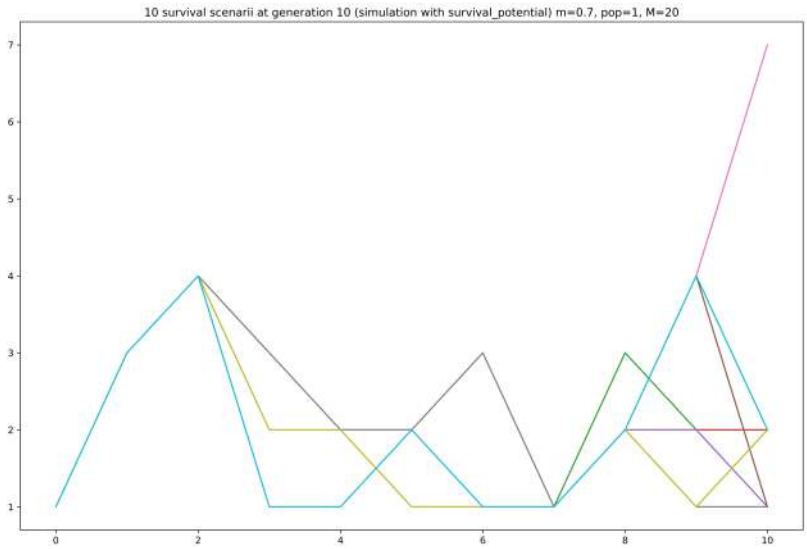

Figure 38: strong imbreeding
The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$ Simulation parameter : $M = 20$

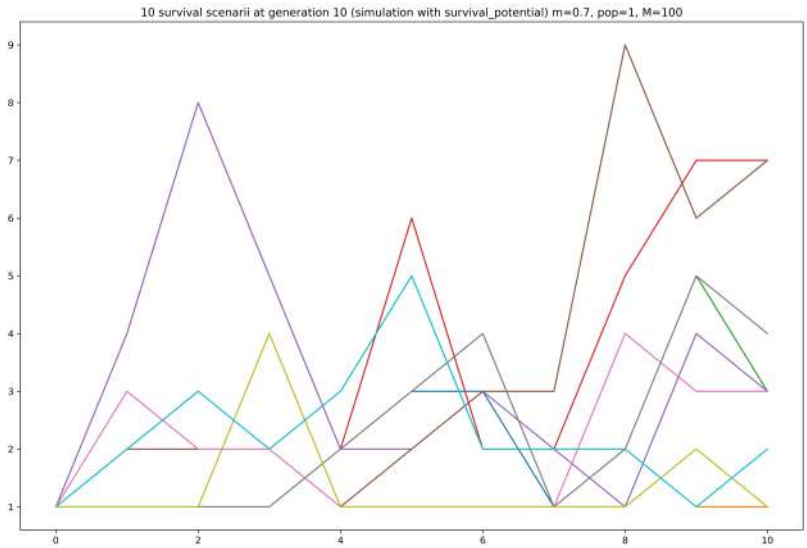

Figure 39: imbreeding
The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$ Simulation parameter : $M = 100$

Imbreeding is characterized by the fact that the trajectories tend to share the same beginning. They are therefore very correlated. It happens because during the selection process, we possibly

extinguish the offspring of a beginning of trajectory. So if we do it many times (n big) and if we have a small number of particles (M small), then two randomly chosen trajectories have a higher-than-theoretically probability to have the same beginning. Of course, the more selective the potential, the bigger M has to be to avoid this problem.

Comparison with naive simulations We simulated survival scenarii both with interacting particle systems using `survival potential` and in a naive way to be sure that our method gives correct results. The following figures show that it is the case.

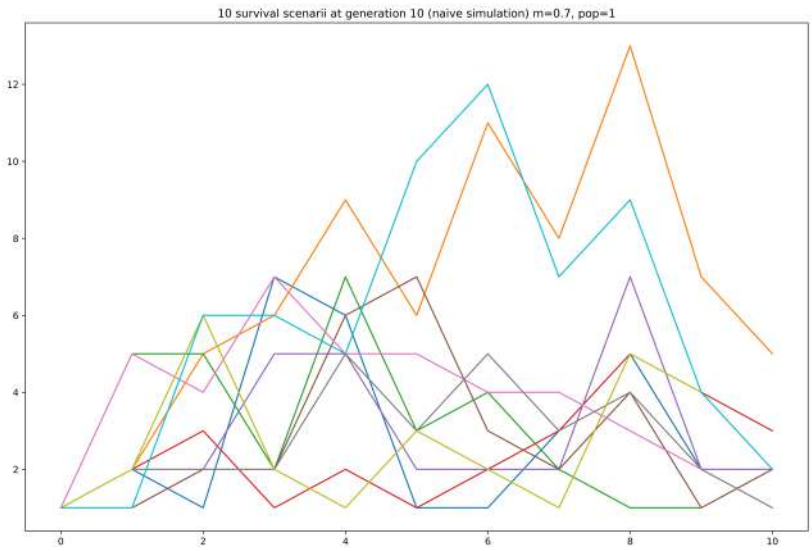

Figure 40: 10 survival scenarii (naive simulation)
The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$

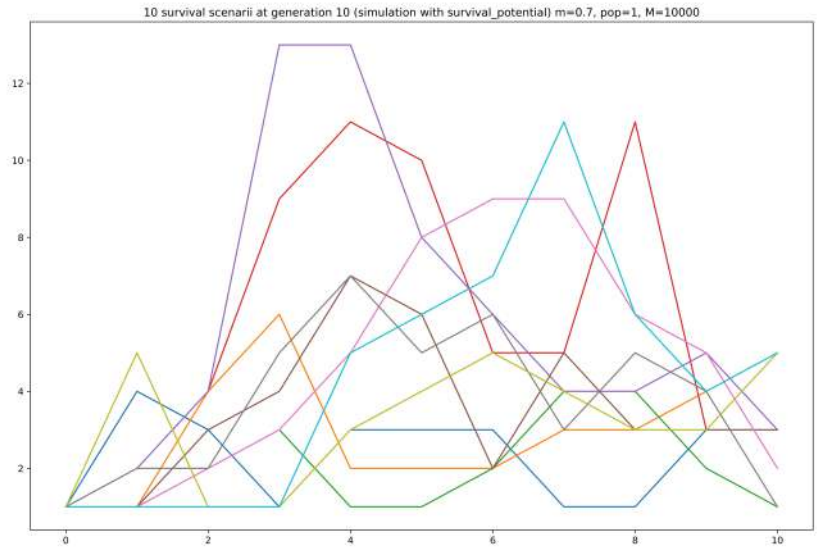

Figure 41: 10 survival scenarii simulated with `survival potential`
The problem parameters are $n = 10$, $pop = 1$, $m = 0.7$ Simulation parameter : $M = 10000$

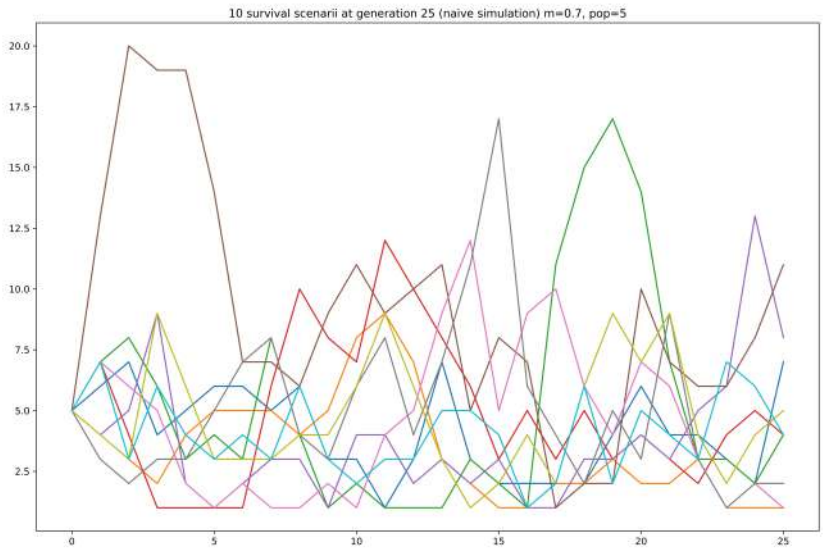

Figure 42: 10 survival scenarii (naive simulation)
The problem parameters are $n = 25$, $pop = 5$, $m = 0.7$

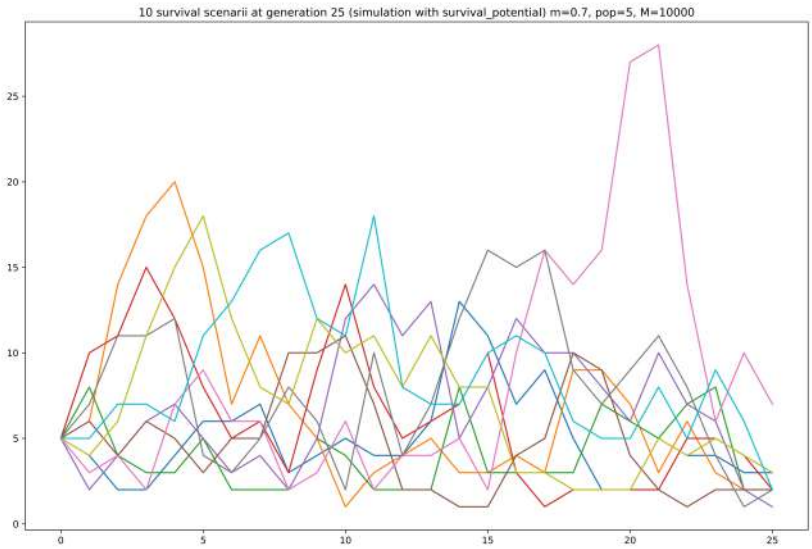

Figure 43: 10 survival scenarii simulated with `survival potential`
The problem parameters are $n = 25$, $pop = 5$, $m = 0.7$. Simulation parameter : $M = 10000$

Study of the influence of the parameters Now we can use our simulator of scenarii to see what happens in more extreme case. We observe mainly four things :

- Typical survival scenarii are given by trajectories that oscillate randomly around a state of equilibrium. Of course they never reach the value 0, but they always come back regularly to values very close to 0.
- Conditional to survival at generation n, the distribution of X_k seem to be approximately the same for all the integers k not too close to O or n. The state of equilibrium is roughly the expected value of this conditional distribution.
- Impact of parameter m. The bigger m, the bigger is this state of equilibrium. In a sense there is "less gravity" when we approach the critic case ($m = 1$), and the "period" of the oscillations also grows with m. The conditional distribution of X_k discussed above is also impacted by m
- *pop* has no long-term influence on the trajectories

The three following figures show it, where we change the value of the parameter m. The state of equilibrium is approximately 3 for $m = 0.4$, 5 for $m = 0.7$ and 10 for $m = 0.9$.

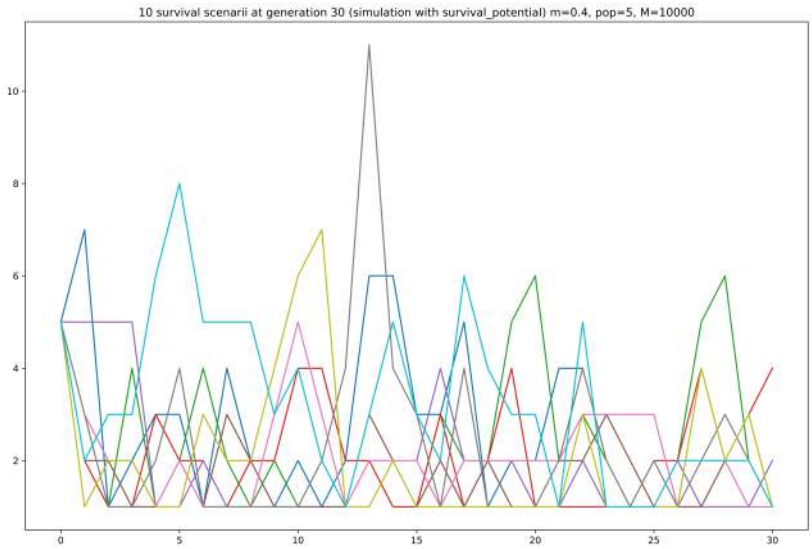

Figure 44: 10 survival scenarii simulated with `survival potential`
The problem parameters are $n = 25$, $pop = 5$, $m = 0.4$. Simulation parameter : $M = 10000$

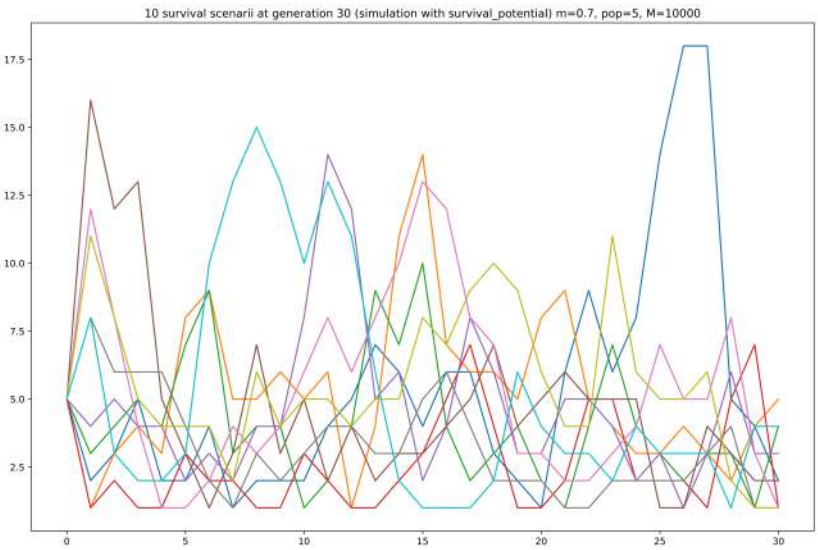

Figure 45: 10 survival scenarii simulated with `survival potential`
The problem parameters are $n = 25$, $pop = 5$, $m = 0.7$. Simulation parameter : $M = 10000$

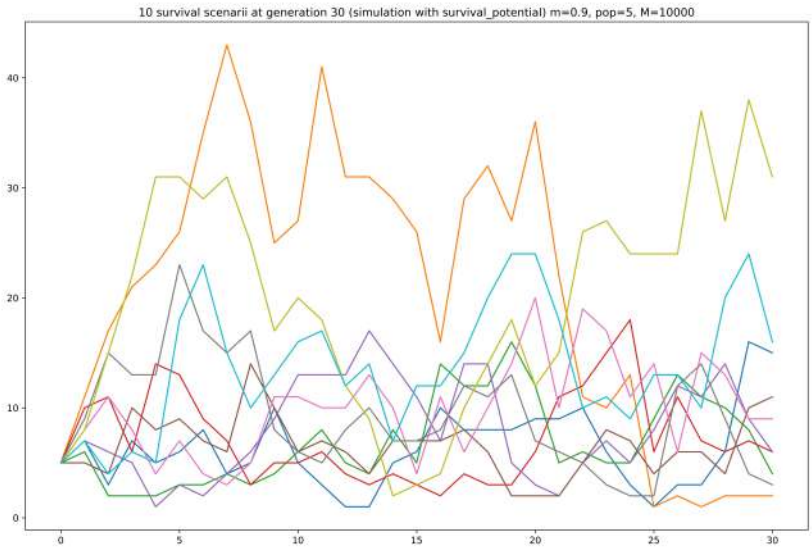

Figure 46: 10 survival scenarii simulated with `survival potential`
The problem parameters are $n = 25$, $pop = 5$, $m = 0.9$. Simulation parameter : $M = 10000$

3.5 Around undergrowth in over-critic case

Over-critic case if defined by $m > 1$. In this case, the undergrowth ($0 < X_n < c^n$ for a constant $c < m$) becomes rare when n grows.

3.5.1 Choosing the best potential

We have the same approach as in the previous subsection (on survival in under-critic case) : we need first to find a potential appropriate to undergrowth.

Here is the list of the potentials we will compare :

- `inverse potential` : $G_p(X_{0:p}) = \frac{1}{X_p}$ if $X_p > 0$, 0 if $X_p = 0$
- `exponentially low potential` : $G_p(X_{0:p}) = e^{-\lambda X_p}$ if $X_p > 0$, 0 if $X_p = 0$
- `log potential` : $G_p(X_{0:p}) = 0$ if $X_p = 0$, $\frac{1}{2}$ if $1 \leq X_p \leq 7$, $\frac{1}{log(X_p)}$ otherwise
- `log bounded potential` : same as above, but the constant b is a lower bound of the potential when n tends to infinity.
- `log log potential` : $G_p(X_{0:p}) = 0$ if $X_p = 0$, 1 if $1 \leq X_p \leq 15$, $\frac{1}{log(log(X_p))}$ otherwise

Note : the apparently strange choices of constant for the three last potentials are just here to avoid indefinite or unmeaning values.

A few words on why we thought to look at those potentials. First, we wanted to have a potential with the following properties :

- $G_p(0) = 0$, because undergrowth excludes dead trajectories
- a potential that favor "low" trajectories, meaning G_p decreasing from rank 1

In the light of those two criteria, we thought to the first two potentials : `inverse potential` and `exponentially low potential`. But in reality those potentials were really bad. The variances of the estimators were too high, and the estimators simply don't work for events of probability 10^{-5}. We wonder why. In short, they were too selective, ie they were favoring "low" trajectories too much, as we observed thanks to function `trajectories`. This is because here trajectories grow often exponentially, so the ratio between the potentials of the highest and the lowest trajectories becomes huge and as a consequence the high trajectories are not selected in the selection process.

Then, we were looking for potentials such that $\frac{G_p(m^n)}{G_p(c^n)}$ doesn't tend to 0 when n tends to ∞ for constants $1 < c < m$, and the function logarithm came naturally. The idea was that trajectories with different growth rate should be able to evolve together, meaning that the ratio of their potentials has to stay reasonable. The two remaining potentials are here as an attempt to be even less selective, ie favoring "low" trajectories even less than `log potential`.

We will now focus on the three potentials `log potential`, `log b potential` and `log log potential`.

First, there is no value of parameter b of `log b potential` that is significantly better than $b = 0$ corresponding to `log potential`. The figure and the table below illustrate it.

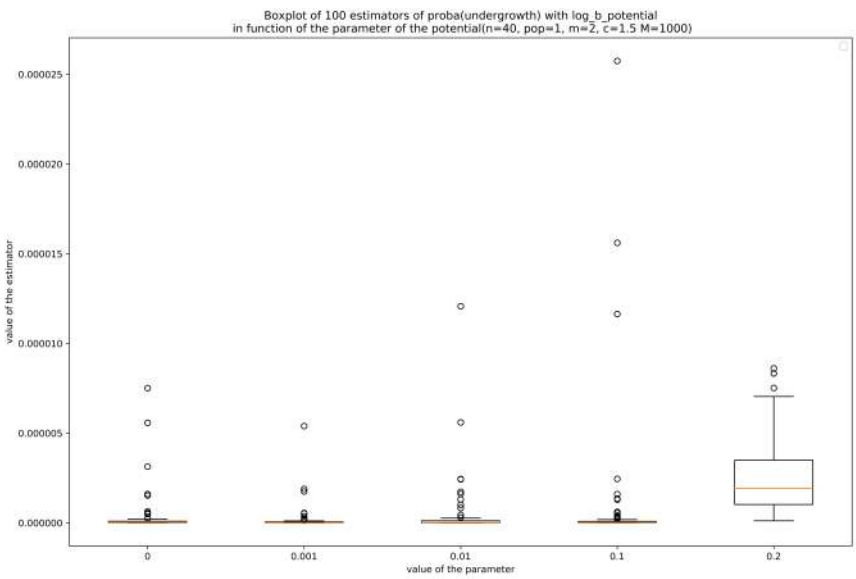

Figure 47: Boxplot of 100 estimators of proba(undergrowth) with `log b potential` for different value of the parameter b

The problem parameters are $n = 40$, $pop = 1$, $m = 2$, $c = 1.5$. Simulation parameter $M = 1000$.

Values of b	0	0.001	0.01	0.1	0.2
Noise-to-signal ratio	3.8	4.0	4.0	4.8	0.8

Table 4: Noise-to-signal ratio with `log b potential` for different values of the parameter b
The problem parameters are $n = 40$, $pop = 1$, $m = 2$, $c = 1.5$. Simulation parameter $M = 1000$.
Noise-to-signal ratio = standard deviation divided by mean. This table has been made with the same realizations of the estimator as the figure above.

Thus, there is no need to consider `log b potential`, which gives no improvement to `log potential`.

Then, there is almost no difference in term of performance between `log potential` and `log log potential`, as you can see below on rare events.

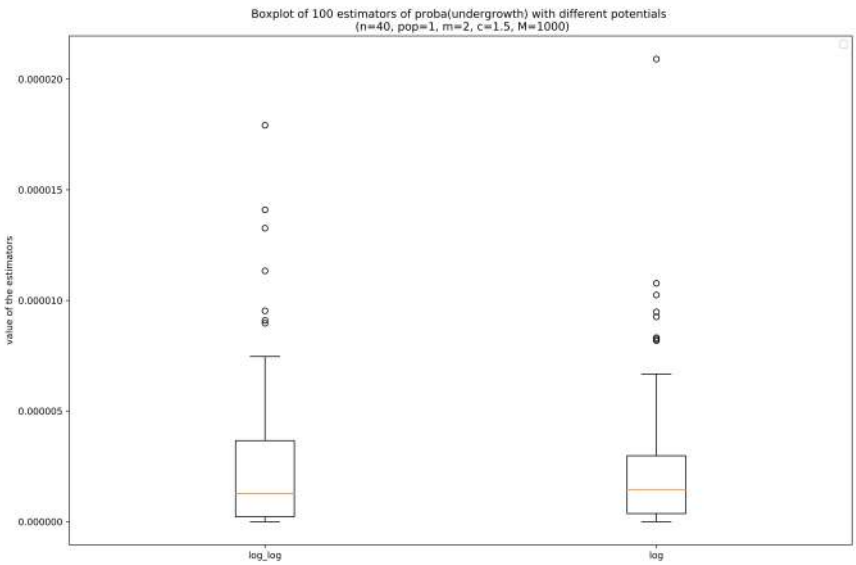

Figure 48: Boxplot of 100 estimators of proba(undergrowth) with `log log potential` and `log potential`
The problem parameters are $n = 40$, $pop = 1$, $m = 2$, $c = 1.5$. Simulation parameter $M = 1000$.

Potential	`log log`	`log`
Noise-to-signal ratio	1.324	1.285

Table 5: Noise-to-signal ratio with `survival potential` for different values of simulation parameter M
The problem parameters are $n = 40$, $pop = 1$, $m = 2$, $c = 1.5$. Simulation parameter $M = 1000$.
Noise-to-signal ratio = standard deviation divided by mean. This table has been made with the same realizations of the estimator as the figure above.

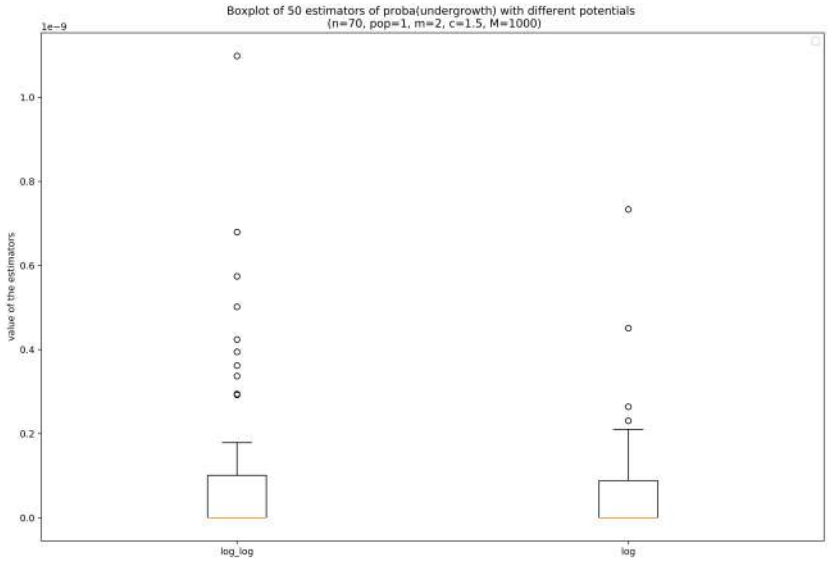

Figure 49: Boxplot of 50 estimators of proba(undergrowth) with `log log potential` and `log potential`

The problem parameters are $n = 70$, $pop = 1$, $m = 2$, $c = 1.5$. Simulation parameter $M = 1000$.

Potential	`log log`	`log`
Noise-to-signal ratio	1.938	2.108

Table 6: Noise-to-signal ratio with `survival potential` for different values of simulation parameter M

The problem parameters are $n = 70$, $pop = 1$, $m = 2$, $c = 1.5$. Simulation parameter $M = 1000$. Noise-to-signal ratio = standard deviation divided by mean. This table has been made with the same realizations of the estimator as the figure above.

Of course, we compared those estimators with a naive Monte-Carlo estimator for not too rare events, and the estimations were the same.

Conclusion of this subsubsection `log potential` and `log log potential` are the two best potentials among those we have studied for undergrowth in over-critic case.

3.5.2 Study of the probability of undergrowth

We haven't found any theoretical results in the literature that we could highlight here. The two figures below suggest that, all other parameters being fixed, the probability of undergrowth is decreasing exponentially to zero when n tends to ∞.

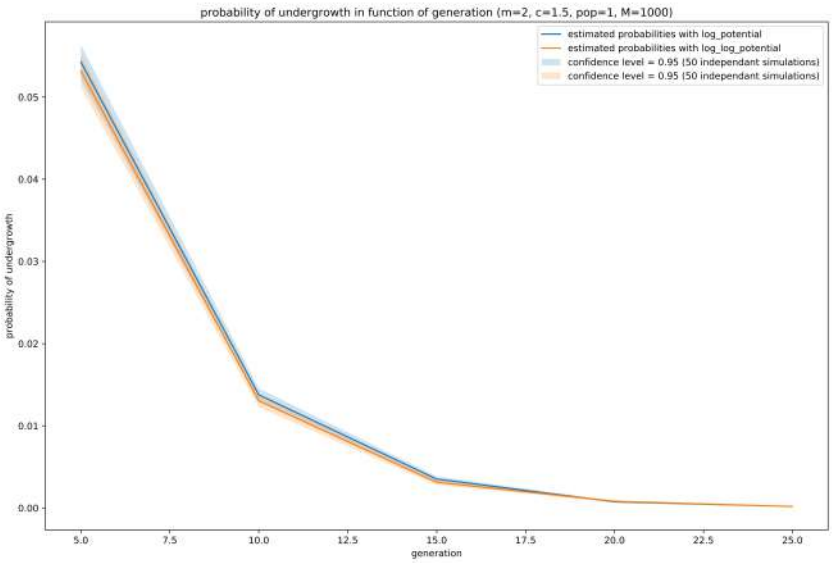

Figure 50: Probability of undergrowth in function of generation (simulation with `log potential` and `log log potential`)

Problem parameters : $m = 2$, $c = 1.5$, $n = 5, 10, 15, 20, 25$, $pop = 1$. Simulation parameter $M = 1000$. 50 independent simulations to build the 0.95 level confidence intervals. Blue = `log potential`, orange = `log log potential`.

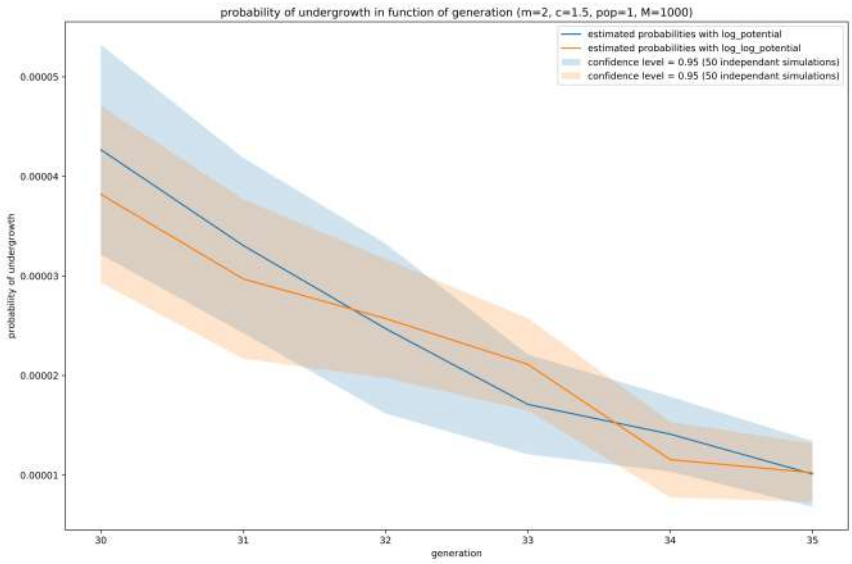

Figure 51: Probability of undergrowth in function of generation (simulation with `log potential` and `log log potential`)
Problem parameters : $m = 2$, $c = 1.5$, $n = 30, 31, 32, 33, 34, 35$, $pop = 1$. Simulation parameter $M = 1000$. 50 independent simulations to build the 0.95 level confidence intervals. Blue = `log potential`, orange = `log log potential`.

3.5.3 Generation of undergrowth scenarii

We use `log potential` and `log log potential` to generate scenarii of undergrowth, as well as a naive generator to be sure that our simulations are correct. The two following figures show that it is the case.

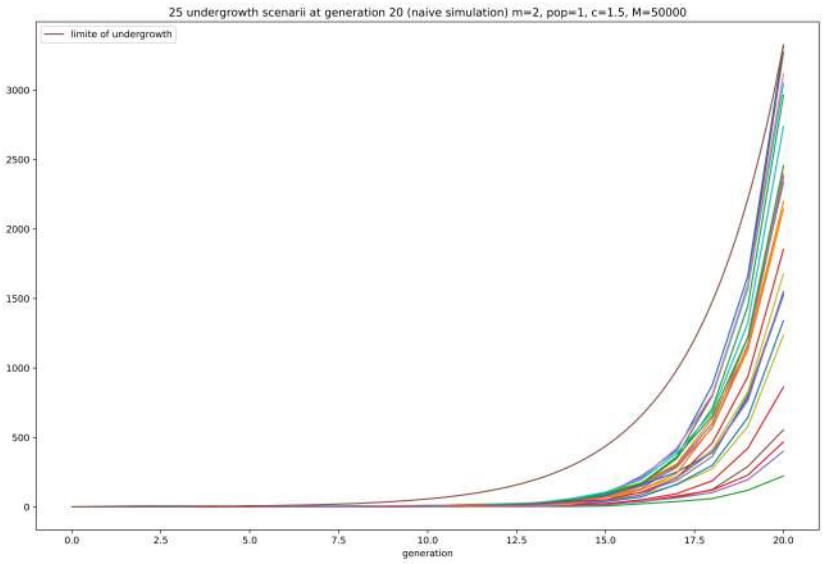

Figure 52: 25 undergrowth scenarii (naive simulation)
Problem parameters : $m = 2$, $c = 1.5$, $n = 20$, $pop = 1$. Simulation parameter $M = 50000$. The curve "limite of undergrowth" is parametrized by (t, c^t)

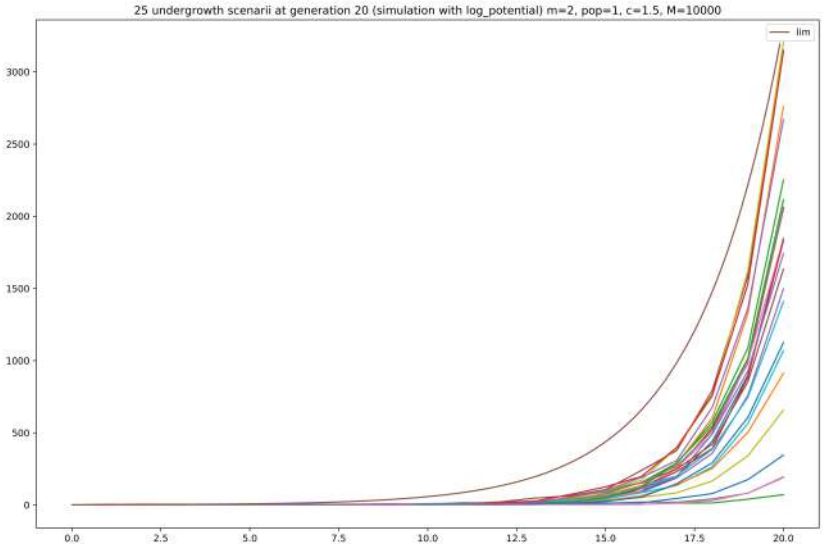

Figure 53: 25 undergrowth scenarii (simulation with `log potential`)
Problem parameters : $m = 2$, $c = 1.5$, $n = 20$, $pop = 1$. Simulation parameter $M = 10000$. The curve "lim" is parametrized by (t, c^t)

Therefore, we can now use our two simulators to look at more extreme scenarii. They give the same results which is of course a good sign (we can't use a naive simulation here because the event is too rare).

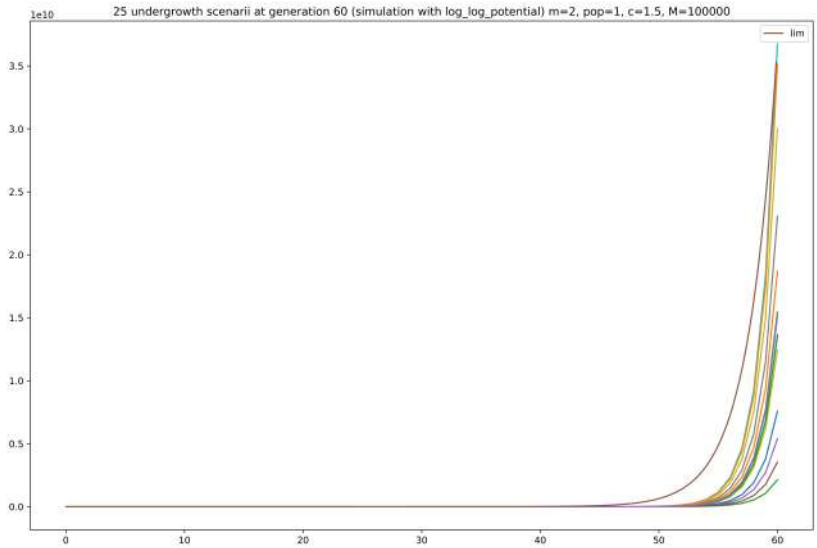

Figure 54: 25 undergrowth scenarii (simulation with `log log potential`))
Problem parameters : $m = 2$, $c = 1.5$, $n = 60$, $pop = 1$. Simulation parameter $M = 10000$. The curve "lim" is parametrized by (t, c^t)

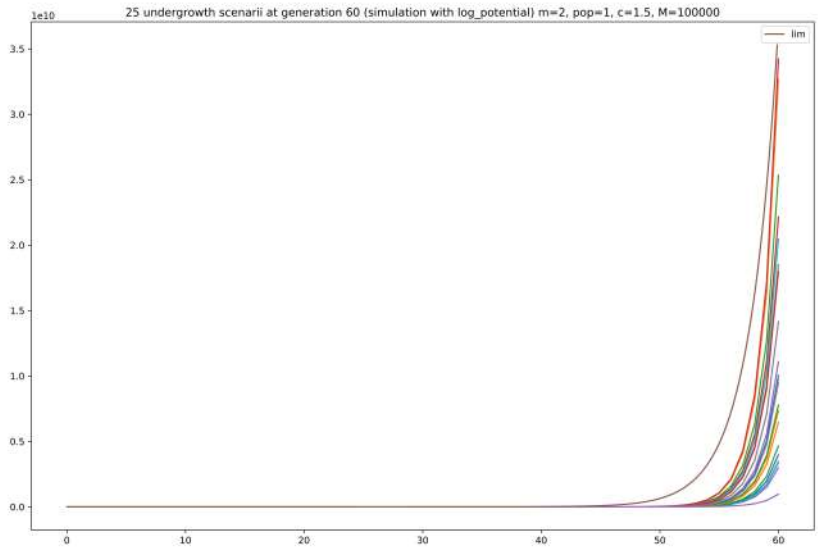

Figure 55: 25 undergrowth scenarii (simulation with `log potential`))
Problem parameters : $m = 2$, $c = 1.5$, $n = 60$, $pop = 1$. Simulation parameter $M = 10000$. The curve "lim" is parametrized by (t, c^t)

From those simulations we can observe that **to achieve a lower-than-normal rate of growth, the process first refrains from branching for a very long time until it can start to grow at the normal rate on the very last generations.**

3.6 Galton-Watson process with two environments

In this subsection, we will look at survival in under-critic case and undergrowth in over-critic case of a bi-environments Galton-Walton process. More precisely, we will focus on realizations of those rare events, and in particular the corresponding sequence of environments.

Intuitively, to achieve survival in under-critic case, it is helpful to evolve in a sequence of environments where the favorable one is over-present, thus we can conjecture that conditional to survival, the proportion of favorable environments in which the trajectories have evolved is on average greater than p_{fav}. We will take a look at this conjecture among others here.

3.6.1 Survival in under-critic case

We begin by simulating scenarii both in a naive way and using interacting particle systems (with `survival potential`) to check that we obtain the same thing. See the six figures below.

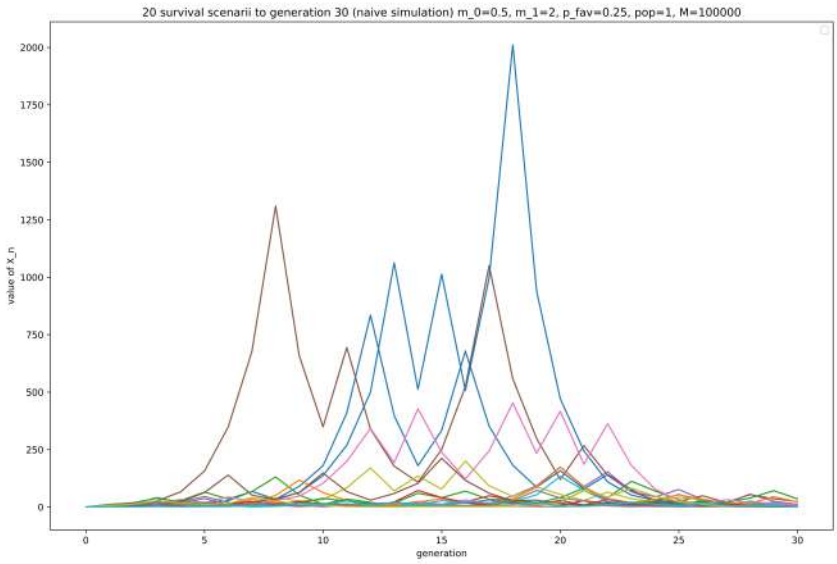

Figure 56: 20 scenarii of survival in under-critic case (naive simulation)
Problem parameters $m_0 = 0.5$, $m_1 = 2$, $p_{fav} = 0.25$, $n = 30$, $pop = 1$. Simulation parameter $M = 100000$

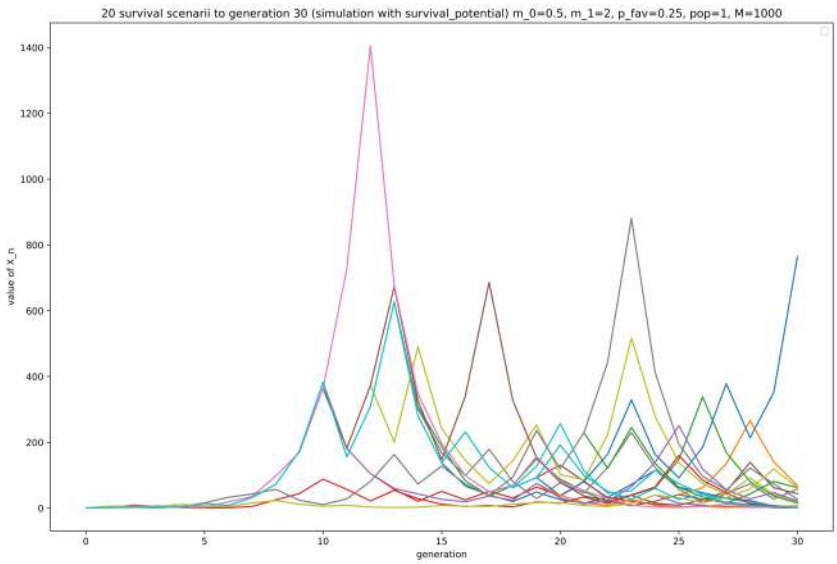

Figure 57: 20 scenarii of survival in under-critic case (simulation with `survival potential`)
Problem parameters $m_0 = 0.5$, $m_1 = 2$, $p_{fav} = 0.25$, $n = 30$, $pop = 1$. Simulation parameter $M = 1000$

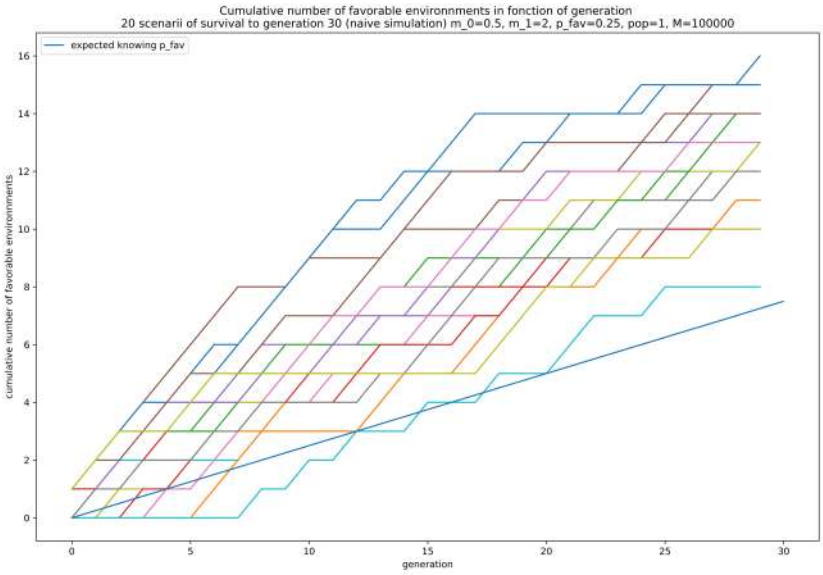

Figure 58: 20 scenarii of evolution of the cumulative number of favorable environments when survival in under-critic case (naive simulation)

Problem parameters $m_0 = 0.5$, $m_1 = 2$, $p_{fav} = 0.25$, $n = 30$, $pop = 1$. Simulation parameter $M = 100000$

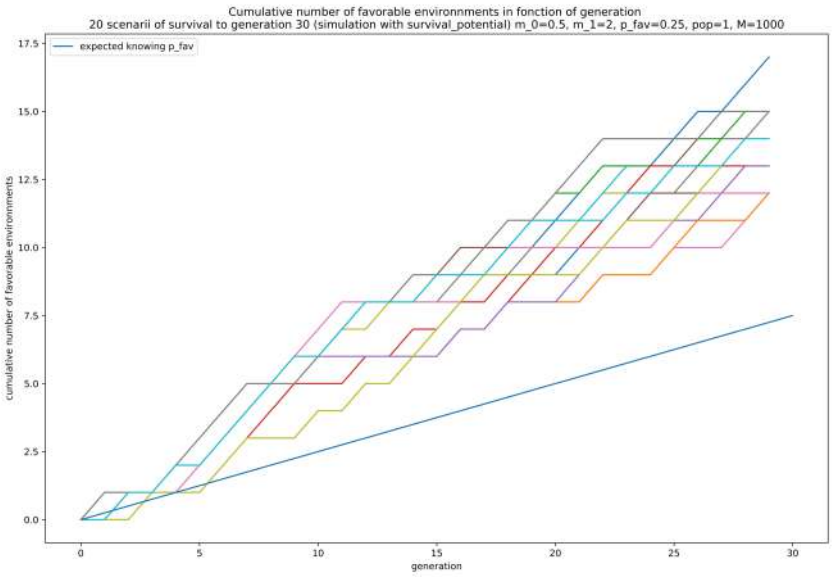

Figure 59: 20 scenarii of evolution of the cumulative number of favorable environments when survival in under-critic case (simulation with `survival potential`)

Problem parameters $m_0 = 0.5$, $m_1 = 2$, $p_{fav} = 0.25$, $n = 30$, $pop = 1$. Simulation parameter $M = 1000$

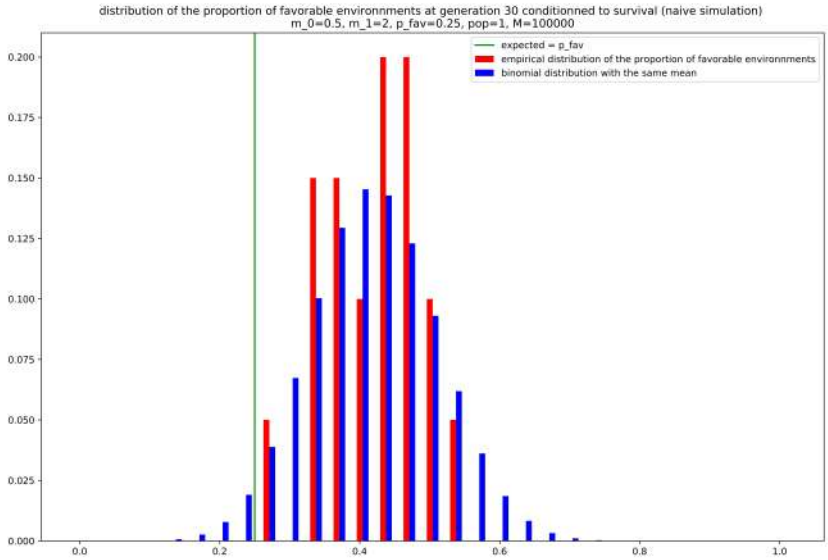

Figure 60: Proportion of favorable environments when survival in under-critic case (naive simulation)
Problem parameters $m_0 = 0.5$, $m_1 = 2$, $p_{fav} = 0.25$, $n = 30$, $pop = 1$. Simulation parameter $M = 100000$

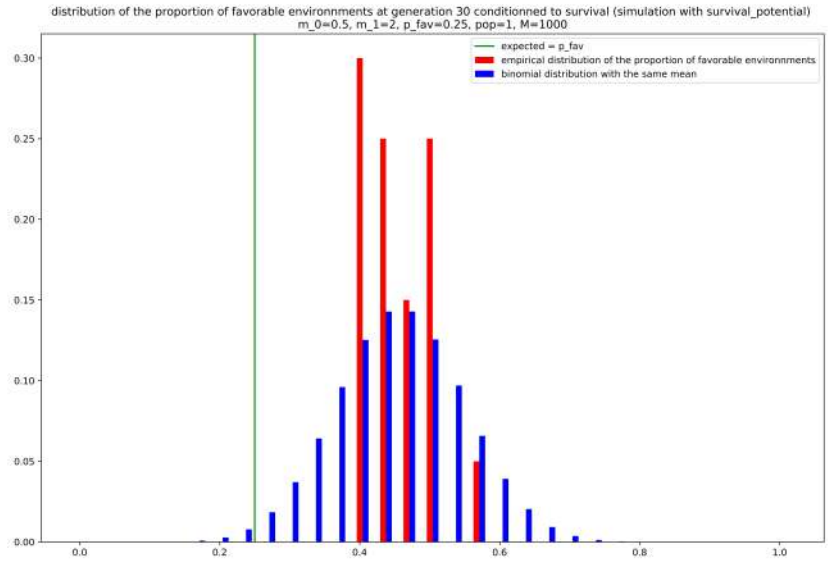

Figure 61: Proportion of favorable environments when survival in under-critic case (simulation with `survival potential`)
Problem parameters $m_0 = 0.5$, $m_1 = 2$, $p_{fav} = 0.25$, $n = 30$, $pop = 1$. Simulation parameter $M = 1000$

Here is what we can see from those figures :

- First, our simulations with `survival potential` seem correct since identical with the naive ones
- On the shape of the trajectories that survive : it is really different from the single environment case. Here we observe very high and sharp peaks. Trajectories regularly approach the value 0, which was also the case in the single environment case, and it is intuitive since we are in a under-critic situation
- As we conjectured, we have $p_{emp} > p_{fav}$ where p_{emp} is the empirical probability of evolving in an favorable environment at a given generation, conditional to survival.
- The favorable environment is over-represented overall but appears in an apparently random order (not specifically at the beginning or at the end of the trajectories)
- The distribution of the proportion of favorable environments over the trajectories is not a binomial distribution. This means that conditional to survival, the environments at each generation are not independent random variables. When looking more carefully at Fig 58 and 59, it seems that environments alternate more often than they would if they were independent

Overall, **the sequences of environments that favor survival the most are those where the favorable environment is over-represented and, in the meantime, where long sequences of consecutive unfavorable environments are avoided.**

When looking at other figures with other values for the problem parameters, we observe that

- p_{fav} being fixed, p_{emp} is decreasing when m_0 or m_1 grows. Intuitively, when the reproduction law are more favorable to survival, the help of a better-than-expected sequence of environments is less helpful.
- m_0 and m_1 being fixed, p_{emp} is increasing in p_{fav} but $\frac{p_{emp}}{p_{fav}}$ is decreasing in p_{fav}.

3.6.2 Undergrowth in over-critic case

We begin by simulating scenarii both in a naive way and using interacting particle systems (with `log log potential`, but with `log potential` we obtain exactly the same thing) to check whether we obtain the same figures or not. See the six figures below.

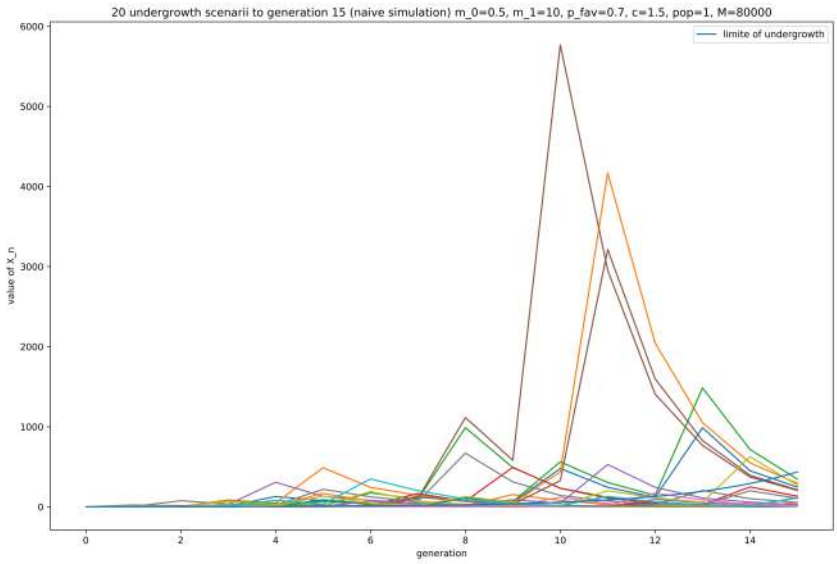

Figure 62: 20 undergrowth scenarii in over-critic case (naive simulation)
Problem parameters $m_0 = 0.5$, $m_1 = 10$, $p_{fav} = 0.7$, $n = 15$, $pop = 1$. Simulation parameter $M = 80000$

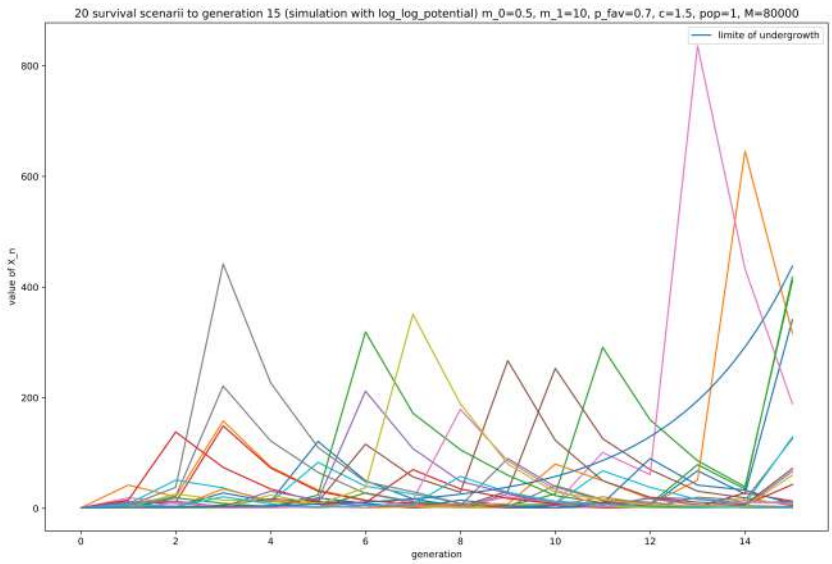

Figure 63: 20 undergrowth scenarii in over-critic case (naive simulation) (simulation with `log log potential`)
Problem parameters $m_0 = 0.5$, $m_1 = 10$, $p_{fav} = 0.7$, $n = 15$, $pop = 1$. Simulation parameter $M = 80000$

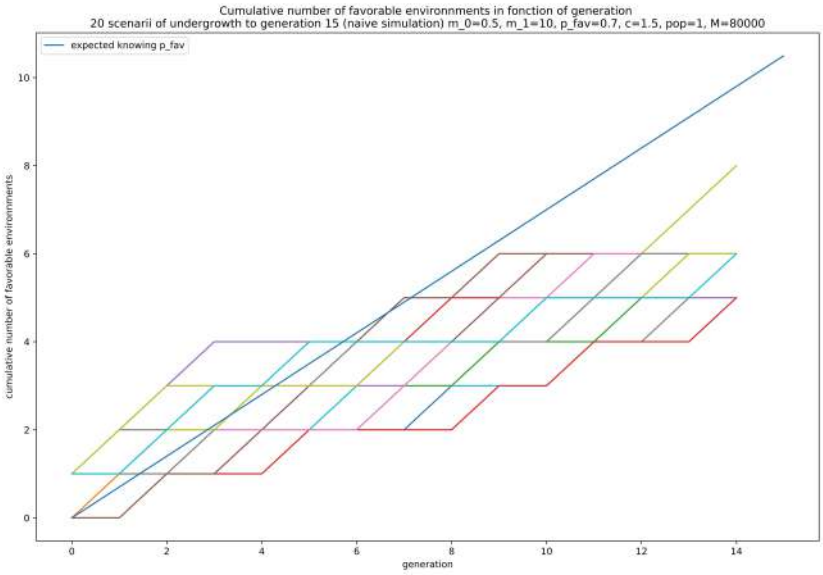

Figure 64: 20 scenarii of evolution of the cumulative number of favorable environments when undergrowth in over-critic case (naive simulation)

Problem parameters $m_0 = 0.5$, $m_1 = 10$, $p_{fav} = 0.7$, $n = 15$, $pop = 1$. Simulation parameter $M = 80000$

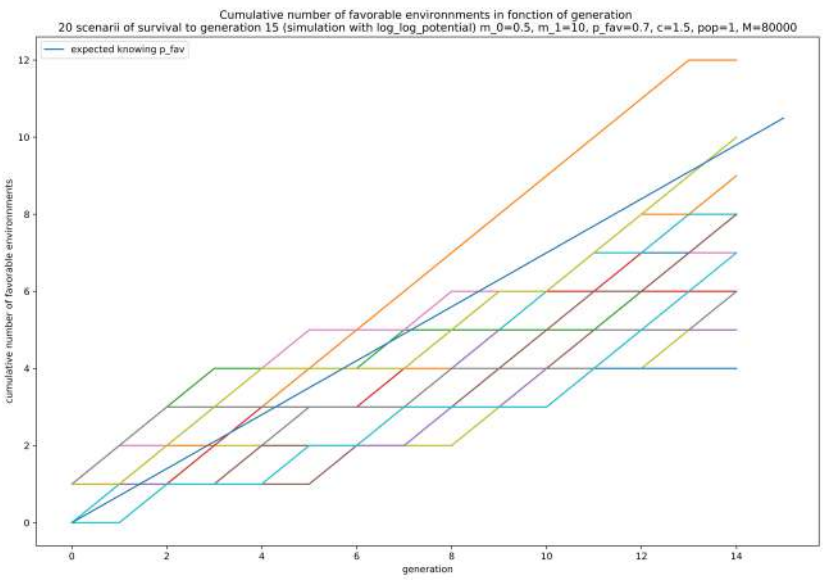

Figure 65: 20 scenarii of evolution of the cumulative number of favorable environments when undergrowth in over-critic case (simulation with `log log potential`)

Problem parameters $m_0 = 0.5$, $m_1 = 10$, $p_{fav} = 0.7$, $n = 15$, $pop = 1$. Simulation parameter $M = 80000$

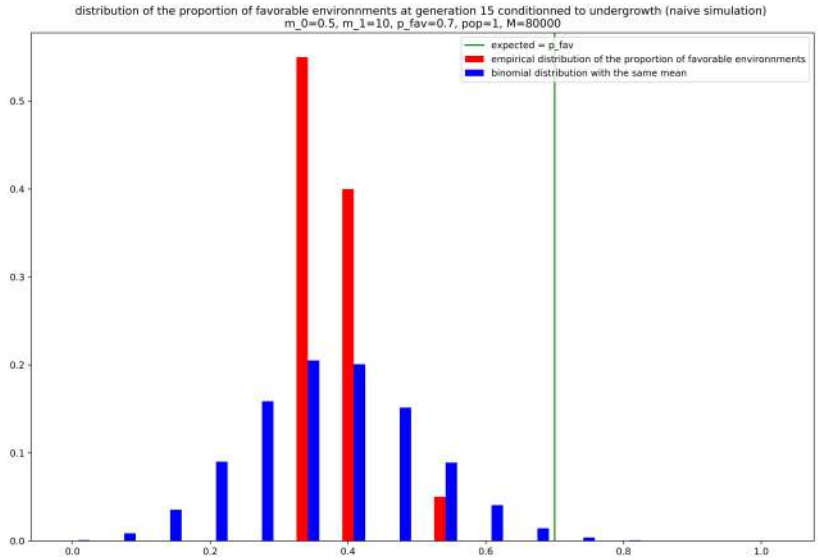

Figure 66: Proportion of favorable environments when undergrowth in over-critic case (naive simulation)

Problem parameters $m_0 = 0.5$, $m_1 = 10$, $p_{fav} = 0.7$, $n = 15$, $pop = 1$. Simulation parameter $M = 80000$

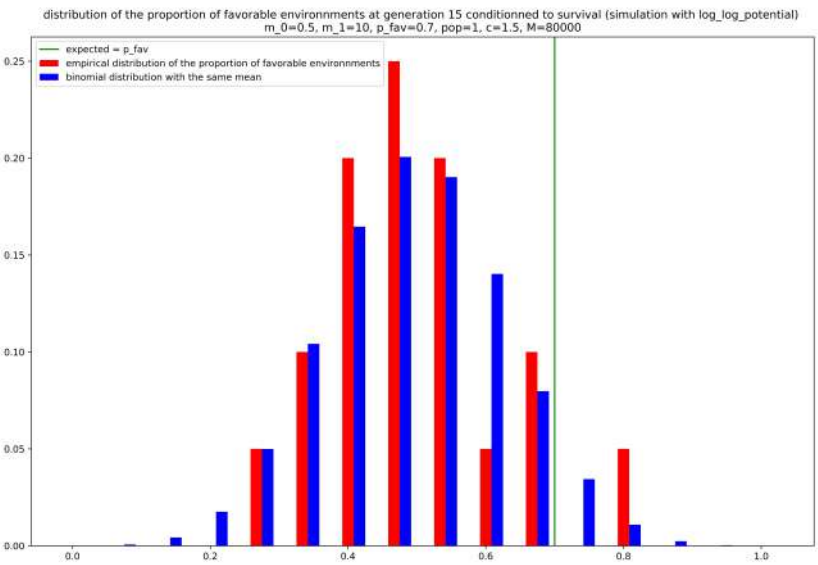

Figure 67: Proportion of favorable environments when undergrowth in over-critic case (simulation with `log log potential`)

Problem parameters $m_0 = 0.5$, $m_1 = 10$, $p_{fav} = 0.7$, $n = 15$, $pop = 1$. Simulation parameter $M = 80000$

Here is what we can see from those figures :

- First, our simulations with `log log potential` are not correct : "high" trajectories are under represented compared to what we obtain with a naive simulation, even with a large number M of simulated trajectories
- On the shape of the trajectories in undergrowth : it is completely different from the single environment case. Here we observe peaks, and trajectories that evolve above the c^t limit to finally satisfy the undergrowth condition $0 < X_n \leq c^n$.
- As we conjectured, we have $p_{emp} < p_{fav}$ where p_{emp} is the empirical probability of being in a favorable environment at a given generation, conditional to undergrowth.
- the favorable environments are under represented but appear in an apparently random order (not specifically at the beginning or at the end of the trajectories)
- the distribution of the proportion of favorable environments over the trajectories is not a binomial distribution. This means that conditional to survival, the environments at each generation are not independent random variables. When looking more carefully at Fig 64 and 65, it seems that environments alternate more often than they would if they were independent

Overall, **the sequences of environments that favor undergrowth the most are those where the unfavorable environment is over-represented and, in the meantime, where long sequences of consecutive favorable environments are avoided.**

When looking at other figures with other values for the problem parameters, we observe that when m_0 increases the peaks become less high. The following figure shows it.

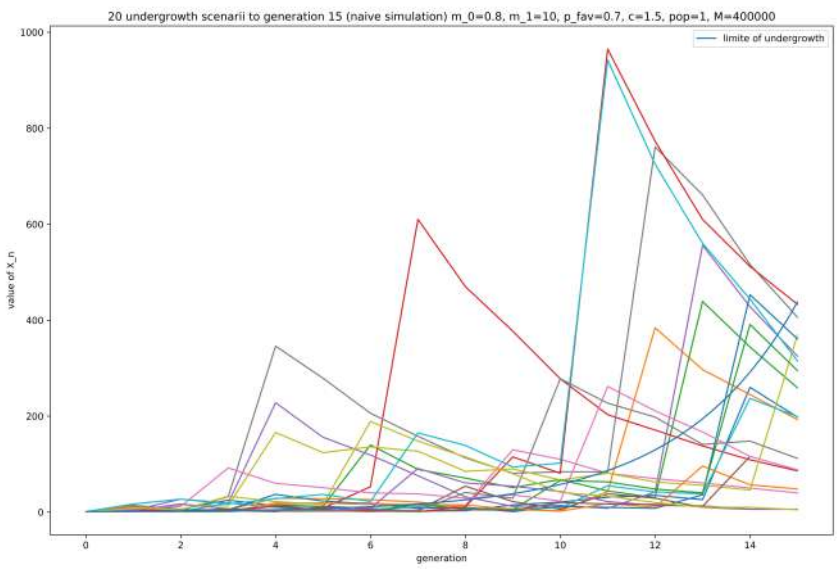

Figure 68: Proportion of favorable environments when undergrowth in over-critic case (naive simulation)

Problem parameters $m_0 = 0.8$, $m_1 = 10$, $p_{fav} = 0.7$, $n = 15$, $pop = 1$. Simulation parameter $M = 400000$

3.6.3 Conclusion of this subsection

We can draw a parallel between survival in under-critic case and undergrowth in over-critic case. The two events are rare ones for big values of n. To achieve them, the environments have a role to play. Indeed, conditional to one of those two events, the "helpful" environment (ie. favorable for survival, unfavorable for undergrowth) is over-represented and in the meantime long sequences of the other environment are avoided.

Finally, the typical trajectories have some high and shark peaks. It was not the case for the classical single environment G.W. process.

3.7 Assessment of this approach

Interacting Particle systems was a very natural technique to be tried on our subject. It was very useful for the generation of typical scenarii.

For survival, the use of `survival potential` was very efficient. Indeed, the estimator was of very good quality : a noise-to-signal ratio of approximately 0.5 for an event of probability 10^{-14} with only $M = 100$ particles on Fig 24 for example. Moreover, it enabled us to simulate survival scenarii correctly and very quickly.

For undergrowth, things are not as good. Granted, the use of `log potential` and `log log potential` gave estimators able to estimate a probability of 10^{-10} with a noise-to-signal ratio of approximately 2 (Fig 49, $M = 1000$). But for the generation of undergrowth scenarii in a over-critic bi-environment Galton-Watson process, high trajectories were under-represented. And of course, it is useless to have a high-performance scenarii generator if it is biased. The main problem is that `log potential` is too discriminating against high trajectories, and attempts to fix it like `log log potential` and `log b potential` don't give better results.

4
COMPARING BETWEEN SIMULATION METHODS

We start our comparison by some indicative values of the time taken by each method to estimate the probability of survival on a fixed noise/signal ratio level. First, let us recall that for the naive Monte Carlo estimator, the runtime is proportional to the number of samples. As an example the effect of dividing the targeted probability by a factor of a 1000, should multiply the number of samples and thus the run time by the same amount. [JL03]

method	`Naive Monte Carlo`	`Importance Sampling`	`Interacting particles`
Runtime *(s)* for m=0.7 Noise/signal≃0.1	0.6	0.5	0.1
Runtime *(s)* for m=0.3 Noise/signal ≃0.1	80.5	0.4	0.1
Runtime *(s)* for m=0.2 Noise/signal ≃0.1	700	0.8	0.2

Both the Importance sampling and Interacting particles are good estimators. The importance sampling should however not be used to estimate frequent events, because of the time taken to compute the negative binomial mass distribution function. The interacting particles has quasi constant time regardless of the rarity of the event and becomes more so if the rarity stems from a big n (number of generation) than a small m. Those main properties of the estimators remain the same even if we change the event we estimate. For instance their performances stay the same for the probability of exceeding x.

The process of evaluating such estimators is rather challenging since the standard deviation is also random and subject to noise, which means that we should iterate the process enough so that we can be sure of our estimates of the std.

REFERENCES

[PE 72] K.B Athreya P.E Ney. “Branching Processes.” In: (1972).

[Ner77] O. Nerman. “On the maximal generation size of a non-critical Galton-Watson process.” In: (1977), pp. 131–135.

[JL03] G. Kersting J. Geiger and V. Vatutin. L. “Limit theorems for subcriticalbranching processes in random environment.” In: (July 2003), pp. 593–620.

[PDe05] P.DelMoralandJ.Garnier. “Genealogicalparticleanalysisofrareevents.AnnalsofAppliedProbability”. In: (2005).

[Mél] S. Méléard. *Modèles aléatoires en Ecologie et Evolution.* Ecole Polytechnique.